Observers Are All You Need

Records, Quantum Identities, Thermodynamics and Closure Equations for Continuing Observers

Bernhard Mueller[1*] Mario Poneder Alexander Osika[2] Kai Xue Ben Cassie
Peter Nguyen Daniel Ari Friedman[3,4]

November 4, 2026

Paper release: r2061 **Released:** November 4, 2026

Author affiliations: [1]Pragma Research Inc.; [2]EtherWorks; [3]Active Inference Institute; [4]FractiAI.

Abstract

This paper develops the observer layer of Observer Patch Holography (OPH). Its hypothesis is that observers are continuing native processes of the executable carrier: each has an accessible algebra and state, retains records in a commutative public record algebra, reads them back, acts on them through record-controlled instruments and continues through checkpoints. At one finite regulator a machine-checked receipt inhabits all seven observer clauses. On finite matrix algebras, every nonnegative normalized valuation additive on coexisting effects is the Born weight of a unique density operator, every pointwise-continuous automorphism flow of a full private block is Hamiltonian, and on the regional-net interface no full matrix factor sits above a scalar overlap region. Given a registered repair law, a declared clock and energy calibration and a refinement-uniform gap family, the four laws of thermodynamics, fluctuation relations and the Landauer bound form one conditional theorem, and the mean entropy production of one repair step equals the relative entropy from its input to the repaired image. The local closure $P = \varphi + \sqrt{\pi}/A_T(P)$ has interval-certified unique roots for each declared map, and the global closure $N = \log M_0(\mathfrak{U}_N)$ reads exact public-record capacity as the independence number of a compound confusability graph; a bounded counterfamily shows that its finite base structure does not fix a unique zero. A supplied charged-scalar/Maxwell action gives first-order convergent real continuum trajectories and a self-adjoint interacting Hamiltonian on its fixed mesh, read out by exact classical record feedback.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation

*Corresponding authors: Bernhard Mueller, bernhard@floatingpragma.ai; Alexander Osika, alex@snrgystudios.com.

nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

This paper's hypothesis and place in the tower

This paper is layer 3 of the tower, the executable observer, and it carries the two closure equations that layer 7 tests. Its hypothesis is the following. *Observers are continuing native processes of the executable carrier with accessible algebras, retained records and record-controlled feedback. The operational quantum identities (Born weights, Lüders conditioning, no-cloning, the Tsirelson bound and Hamiltonian form of continuous symmetry flows) and the thermodynamic identities (second law, Clausius inequality, fluctuation relations, Landauer bound and finite-regulator third law) follow from that observer structure once one repair law and one shared reference are supplied on the same carrier.*

The unconditional content is a set of finite theorems: the inhabited observerhood receipt and common-origin theorem, the character-block obstruction, the finite Stone converse, the finite effect-valuation theorem, the event-algebra classifications, the finite history-weight and derived-action

theorems, the public-capacity theorems and the bounded capacity counterfamily. The four-law package, the closure readings of P and N and every physical reading of a modular parameter are conditional implications whose premises are displayed where they enter.

The executable model must realize several open joins before these conclusions apply to it: one continuing native observer with every service and resource accounted; a source-produced, complex-tomographically complete effect and instrument web together with an operational derivation of noncontextual additivity; one source reference shared by the state and transition optimizers; the physical clock and energy dictionary; a complete capacity source antecedent with one physical zero of the finite-size slack; and the same-quantity bridge and target-independent map selection for P.

The hypothesis or a branch of it is refuted by a counterexample to a load-bearing finite theorem under its stated premises; by a native law whose records admit no common state and transition reference (the audited direct obstructions below are such results for one artifact); by observables that change with hidden carrier presentation after the visible interface is held fixed; by a source-realized complete effect web on which a non-Born valuation is additive; by a nonunique or target-leaking closure solve; or by the qualifying comparisons stated in the companion papers. Gauge structure, matter, spacetime and gravity are developed in the companion papers [1, 2, 3]; this paper points to them rather than re-deriving them. Appendix A records the finite field histories, on a supplied charged-scalar/Maxwell action, that observer records read and restore.

Keywords: observer patch holography; quantum foundations; observers and records; finite thermodynamics; Born rule; closure equations

Contents

1 Observers, Records and Clocks **5**

2 Observerhood and the Common-Origin Bridge **11**

3 Records and Quantum Identities **14**
3.1 Finite event algebra . . . 14
3.2 The Born weight: effect-valuation theorem and frame boundary . . . 14
3.3 Record algebra, measurement, Bell correlations and checkpoints . . . 16
3.4 Finite regional observer nets . . . 18
3.5 A noncommutative join of the retained observer carriers . . . 18

4 Common-World Joins on the Twelve-Port Carrier **20**

5 Histories, Actions and Mechanics **26**

6 Thermodynamics from Conditional Repair **35**
6.1 Protected records with decaying repair memory . . . 38
6.2 Elementary corollaries of the record and screen structure . . . 43

7 Closure read by the same observers **45**
7.1 Pixel ratio, currents and mass response . . . 45
7.2 Physical scale and calibration ancestry . . . 46
7.3 Global closure for total capacity . . . 46
7.4 What closure establishes . . . 51

A Finite Field Histories Read by Observers **53**
A.1 A finite interacting field and its continuum boundary 54

B State-and-Law Habitat **54**

C Observer Registry and Clock-Naturality Certificate **57**

1 Observers, Records and Clocks

An abstract observer patch is the algebraic object

$$\mathsf{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \mathrm{Chk}_i)\,.$$

Here $\mathcal{A}_i$ is the finite or regulated accessible algebra, ρ_i the accessible state, $\mathcal{R}_i$ the exact or approximate record algebra, $\mathcal{I}_e$ the overlap-visible interface algebra, $\pi_{i,e}$ the visible restriction, $\mathcal{U}_i$ the allowed update and repair instruments, and Chk_i the checkpoint data used for continuation.

A support patch, such as a cap $P_i \subset S^2$ or a causal diamond, is a geometric chart of the abstract patch on a geometric branch. A carrier patch is a physical or digital implementation that realizes the same visible interface and record statistics within a declared error. Echosahedral hardware is therefore a reference carrier architecture for a fixed-cutoff implementation surface. The physical claim lives at the quotient of visible records, interface statistics, repair maps, and checkpoint continuation.

The quotient removes presentation data such as coordinates, labels, worker layout, and hidden ancillas. Quotient-visible port number, oriented incidence, topology, and response maps are physical branch data when the observer can recover them. Implementation invariance therefore does not imply carrier neutrality.

The observer patch is therefore the formal access structure used by the proof. A human observer, a horizon cap, and a hardware module can realize or chart it, but none of those examples is the definition. Physical predictions attach to quotient data: visible interface statistics, record readout, repair maps, and checkpoint continuation. A carrier implementation is admissible only when those data are invariant within its declared error model.

Definition 1.1 (Semantic observer history)**.** *For an observer token O, the observer-readable history is a labeled informational poset*

$$\mathsf{H}_O = (E_O, \preceq_O, \ell_O).$$

The elements of E_O are semantic record events, $\preceq_O$ is informational dependence, and ℓ_O is the observer-visible event label. The strict part of the order is the transitive closure of certified read-from parenthood, and $\preceq_O$ is its reflexive closure. The strict part is machine-checked as the least strict extension of the direct parent edges. A direct edge requires the child to certify the resource, the parent to write it, the child's pre-commit snapshot to name that parent as writer, and the pre-commit value to equal the parent's post-commit value; a raw writer label alone creates no edge. One adjacent independent swap preserves the authenticated direct-parent relation and its closure only under the fresh-ID and duplicate-free hypotheses on both executions; the general swap-chain theorem preserves only the raw writer-citation relation. Executor stutters are removed before the commit list is formed. Event identity is defined from the canonical semantic payload, observer token, visible footprint, and semantic dependency parents. It is not derived from worker ID, repair iteration, queue position, retry counter, timestamp, or packet latency metadata. This definition is not a theorem that arbitrary schedule rewrites preserve labels or payloads. Latency is execution gauge only when it does not alter informational dependence, delivered data, timeout-sensitive operations, or operational clock records.

The strict order of Definition 1.1 is causal-set-like in a limited sense: Lean proves that reflexive authenticated ancestry is a partial order whose intervals are finite at each cutoff, so the order-theoretic causal-set axioms hold [4, 5]. The relation is informational read-from dependence. Its canonical source height, zero at roots and one plus the maximum direct-parent height otherwise, is exactly the longest authenticated parent-chain length, and every finite strict partial order compiles exactly into an abstract authenticated log. An OPH commit log is not a Poisson sprinkling and

its event count is not a spacetime volume; physical embedding, a volume law, a four-dimensional manifold limit and Lorentz kinematics are developed, with their premises, in the spacetime paper [3]. There a rank-three source Gram quotient and a real axis give a four-dimensional target carrier of inertia $(1, 3)$, source placements become exact finite causal-order embeddings under stated speed and spacelikeness conditions, and conservative record populations with a declared model clock have flat $1 + 3$ order and count-volume limits. Thirteen finite covariance and isometry table identities of the rank-three port-record completion use Lean's `native_decide`, so those subreceipts trust the native compiler and runtime in addition to the kernel and are not kernel-only proofs.

Definition 1.2 (Observer registry groupoid and namespaces)**.** *Observer identities live in a registry groupoid, not in a flat UUID table. Objects are observer tokens*

$$\mathrm{RID}(O) = (\mathrm{kind}(O), [\mathrm{birth}(O)]_{\mathsf{H}}, \mathrm{lineage}(O)),$$

where $\mathrm{kind}(O)$ *lies in the disjoint namespace*

$$\{\textit{patch observer},\ \textit{cap observer},\ \textit{future observer}\}.$$

The birth event is a semantic event class in H*, and lineage is carried by explicit continuation, split, and merge arrows. Local registries on overlapping presentations must preserve observer kind, birth event, lineage arrows, visible observer structure, and checkpoint cuts. The global registry is the descent groupoid or colimit of those local registries when the cocycle and trivial-monodromy checks pass. A split creates child tokens with parent links; it does not silently assign one identity to multiple descendants.*

Definition 1.3 (Operational observer clock)**.** *The modular parameter of an observer-facing cap pair is not, by itself, an arbitrary operational clock reading. A clock claim supplies an observer-accessible instrument*

$$\Theta_O : \mathrm{Chk}_O(D) \to \mathrm{Prob}(\mathbb{R})$$

and a declared affine calibration law. For a corresponding observer O'*, the required finite or exact certificate has the form*

$$\Theta_{O'}(\Phi_{O*}\rho) \approx (A_{a,b})_*\Theta_O(\rho), \qquad A_{a,b}(\tau) = a\tau + b,$$

with an explicit residual bound from state mismatch, calibration error, and record-readout error. Execution counters are provenance; observer record order, modular parameter, and clock uncertainty are separate semantic or operational fields.

Remark 1.4 (Observer-clock naturality boundary)**.** *This paper proves observer-first quotient and readback claims only where the cited theorem surface supplies the required data. Scheduler-independent observer history and clock naturality require a history-augmented quotient relation with history-coherent diamonds, registry descent, state-preserving observer-algebra extraction, support-cap chart naturality, and the operational clock instrument above. A public evidence bundle can falsify the claim by showing counter-dependent semantic event keys, duplicate or namespace-colliding registry identities, broken registry cocycles, non-isomorphic semantic history DAGs across executors, or clock residuals exceeding the declared bound. Implementation field names alone are not the proof.*

The registry, history and clock data above form one finite evidence contract. Theorem C.1 in Appendix C states the conditions under which executor metadata cannot change the observer-readable history class, registry class or clock law.

The formal time-and-order ledger enforces this boundary at the type level. Universe closure, repair execution order, observer record order, modular parameter, worldline realization, clock readout, proper time, and an optional global time function are eight separate structures. In the committed source environment, the complete matrix of 56 ordered transitive coercions between distinct ledger layers fails to elaborate. Interpretations pass through an explicitly named realization map without a function coercion. More strongly, composition with any strictly increasing real function preserves the bare clock interface. A three-record control sends readings (0,1,2) to (0,1,8) and proves that the resulting clock is not related to the original by any positive affine map. Record order therefore supplies order data and an order-compatible scalar readout, while fixing neither affine structure, origin, nor physical unit.

The bounded observer-time continuation keeps the missing receipt visible. If a supplied event atlas and record map satisfy an affine coordinate law along a supplied future-unit timelike direction, every precedence has future-timelike displacement in every overlapping chart. On this calibrated branch the positive clock increment along that same supplied affine history is additive, and its square equals the chart-invariant Lorentz quadratic interval. One shared public event cannot determine affine rate and origin. Two ordered shared events determine the unique positive-affine interpolation of their four readings. At a third shared event, affine consistency is equivalent to an exact cross-multiplication equation. Distinction of the event and both readings from the two anchors makes this a nondegenerate check with no new fit parameter. Calling it held out requires a separate predesignation and custody protocol. The affine and cubic controls synthesize separate atlases from their clocks; they pass and fail this algebraic check without constructing two physical clocks on one calibrated geometry. A separate literal three-record, two-chart example proves the geometric interface is inhabited. These are bounded conditional algebraic theorems with finite controls: the source atlas, affine unit-speed law, shared-event correspondences, refinement transport, physical instrument, common physical clock geometry, predesignation protocol, three-clock network, and SI unit are not constructed, and no global time function or modular-to-clock identity follows.

The companion formal tower interface packages the next structural layer in one object: finite observer and record fibres, observer-indexed record orders, private matrix algebras, commutative public subalgebras, selected states and generators, and functorial refinement maps with explicit compatibility laws. An exact constant adaptor packages one existing projective partition and certified state using a discrete order and zero generator. This is an inhabited packaging witness, not a nonconstant source-derived tower, repair endpoint, causal net, physical evolution, geometry, clock, or continuum limit.

The observer-patch definition does not fix the primitive observer size. A1–A3 require finite or regulated access at fixed cutoff and permit connected support regions with different finite or regulated algebras. A primitive observer may therefore be a variable finite federation of screen cells. The stronger claim that elementary carriers are isomorphic fixed-capacity cells would require a separate branch: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlap maps, refinement preservation of the cell type, and a proof identifying one cell or a fixed block of cells with the primitive carrier. One Echosahedron is therefore a candidate primitive carrier on the homogeneous branch. It is not the definition of an observer patch.

Calibration is an import. The local readout of a selected normal form supplies an internal record history, not a physical clock. Operational time requires an observer-readable transition process, event correspondence and affine calibration; it is not the external update parameter. In the Lean corpus the 2019 SI anchors, the cesium hyperfine frequency, the speed of light and the Planck constant, enter as exact rational literals with a unique inhabitant, so the dictionary itself

carries no measurement uncertainty, and the entire empirical content of the import is one declared tick duration. Under a declared tick the internal step count, angular rate and energy gap convert exactly to laboratory seconds, hertz and joules, the conversions compose affinely with the committed clock algebra, and two distinct tick declarations give distinct laboratory frequencies for the same internal rate, so the calibration is an import and never a derivation.

The premises A1–A3.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0 ,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise registry.

The closure hypothesis. The theorem premises A1–A3 state the library's hypotheses on the carrier and its response, on internal transport and on local state selection. The closure hypothesis states that the Universe is a timeless, self-referential structure whose observers recover the specification of their own world from their records and realize it, so that the description and the

described system are one object. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The closed object is the complete inhabited history, whose internal events and responses to interventions are its proposed intrinsic computation. This reading requires no external computer or playback time. It is distinct from fixedness of a repair state, a public record, or a probability law, and does not itself construct the history or prove a final cosmic endpoint.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point per declared map on its analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as diagnostics of the declared source maps. The latter absolute residual is 3.3904×10^{-4}, compared with experimental standard uncertainty 2.1×10^{-8}; this is not agreement at measured precision. The gauge width supplies 99.2 percent of the gap between the bare root and the measured value; the residual 3.3904×10^{-4} is addressed to same-scheme hadronic transport to the Thomson endpoint.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the A1–A3 source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\!\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\mathrm{pres}} = N_0\left(1 - \frac{P}{24}\right), \qquad \ln\frac{N_{\mathrm{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\mathrm{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122}. The full weighted Planck base-ΛCDM chain gives the comparison coordinate 3.3129×10^{122}. The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

The descent repair law turns compatible local corrections into a schedule-independent public record once termination and the local diamond hold; the consensus paper proves this finite normal-form theorem and its countermodels [6]. Counting and trace give exact measures on the carrier algebra, and agreement and repair decide whether those measures survive on the public quotient. A physical readout must preserve the resulting measure before it can be identified with a laboratory or spacetime quantity. The local gauge Lie type that complete reversible response and endogenous overlap transport force, and the conditional matter image, are proved in the gauge paper [1]; the particle paper carries the matter and mass calculations [2].

2 Observerhood and the Common-Origin Bridge

The observer named in the title has a theorem-level bounded receipt. Over any instance of the companion formal tower interface, its base structure packages clauses 1 through 6 and the own-observer half of clause 7 as typed data: a bounded accessible star subalgebra of the private algebra containing the observer's own public record algebra; a self-readback map sending accessible elements into that public algebra and fixing each of its elements; durability, with every record element annihilated by the observer's declared generator; record-conditioned control maps preserving the accessible interface; a prediction map whose output never strictly precedes its input in the observer's own record order; label and prediction families commuting with the declared refinement maps; and a declared readout equal to the trace pairing of each record element against the observer's own state, with every record element fixed by every declared control. Readback idempotence and refinement stability of the readout follow. An explicit witness inhabits the receipt away from the constant adaptor: private algebra $M_2(\mathbb{C})$, a nonzero generator given by the scaled commutator with $\mathrm{diag}(1,2)$, which annihilates every diagonal record element while moving the raising matrix unit, control by conjugation with the involution $\mathrm{diag}(1,-1)$, three chain-ordered records, and a nonconstant non-identity prediction map. Four negative controls (an erasing readback, an identity generator, a resetting prediction, and a constant misreading) each violate one attained field on the same data. The base structure is operational and finite: it selects no unique observer over a given tower, attaches no physical instrument, makes no consciousness claim, and supplies no source realization of the tower data. A separate fixed-regulator mixin closes the cross-observer half on an access cut. It requires two distinct labels, exact agreement of both accessible interfaces with the cut, equality of every committed record after the two typed restrictions to the owner-region meet, equality of own readouts, and a nonzero common restriction. The common section is thereby accessible to both observers. Its proper-meet witness uses two distinct diagonal owner algebras,

a scalar bottom algebra, and corner-character restrictions: observer-indexed records differ before restriction and agree nontrivially only after restriction. A one-corner mutation makes the overlap receipt false. The operational overlap-event consumer takes two operational observers owning two charts in which one event is visible and proves equality of the two owner-region restrictions of its packet record together with two-sided self-readback. Thus all seven operational clauses are inhabited at one finite cut. Cross-regulator naturality, triple-overlap coherence, and source realization are not supplied [7].

An access cut on the finite regional observer net of Section 3.4 assigns each observer one declared support region and one accessible star subalgebra squeezed between the observer's public record algebra and the regional algebra of that region, stable under the declared restrictions below it; the observer-local algebra at a region is the meet of the regional algebra with the accessible algebra, and the net receipts transport to this meet. A finite event world over the supplied Lorentz soldering binds one packet p_e, with packet carrier fixed independently of the quotient state, to each event e, with declared chart ownership. The common-origin theorem states that for every active visible chart/event pair (i, e),

$$\operatorname{supp}(\mathsf{p}_e) \preceq R_{O(i)}, \qquad \operatorname{rec}(\mathsf{p}_e) \in \mathcal{L}_{O(i)}\big(\operatorname{supp}(\mathsf{p}_e)\big) \cap \mathcal{P}_{O(i)}, \qquad x_i(e) = \operatorname{geo}_i(\mathsf{p}_e),$$

where $O(i)$ is the declared owner of chart i, $R_{O(i)}$ its observer region, $\mathcal{L}_{O(i)}$ its observer-local algebra, and $\mathcal{P}_{O(i)}$ its public record algebra: one event packet determines the event support region below the owner's region, the public record observable in the meet of the observer-local algebra at that support with the owner's public algebra, and the Lorentz chart coordinate as a readout of the same packet. A selected coarse-to-fine refinement transports all three conjuncts. The witness chain is strict, with the public algebra strictly inside the accessible algebra strictly inside the private algebra, and the witness records separate packets. Charts and observers are distinct types, ownership is declared data, and no physical region, coverage, instrument, spacetime, causal cone, event actualization, or continuum object is claimed [7].

Coexistence of regional operator algebras on this net interface has an exact boundary.

Theorem 2.1 (Scalar-character obstruction and character-block coexistence)**.** *For every $n \geq 2$ there is no unital complex-algebra homomorphism from the full matrix algebra $M_n(\mathbb{C})$ into the scalars, and no unital complex-algebra homomorphism into the scalars accepts an anticommuting pair of units. On this net interface, at any regulator of nonzero dimension, a region whose local algebra consists of scalars admits above it no region whose local algebra contains a unital copy of $M_n(\mathbb{C})$ with $n \geq 2$: the declared star-homomorphic restriction retraction would compose with the copy to a scalar-valued unital homomorphism on a full matrix algebra. At any such regulator, a region lying above a region with commutative local algebra carries no anticommuting pair of units: the declared restriction would send the pair to commuting units whose product equals its own negation. No full matrix factor of dimension at least two sits above a scalar overlap region, and the standard tensor-factor local-algebra picture with trivial overlap is impossible on this interface. The boundary is inhabited: over a dimension-four private algebra, the block algebras of matrices* $\operatorname{diag}(a, a, A)$ *and* $\operatorname{diag}(B, b, b)$ *with arbitrary two-by-two blocks A and B are each noncommutative, commute elementwise, neither contains the other, and each retracts onto the scalar overlap algebra through the character evaluating its one-dimensional corner block.*

Proof. The matrix ring $M_n(\mathbb{C})$ is simple, so the kernel of a unital homomorphism into the scalars is trivial and the homomorphism is injective, while the off-diagonal matrix unit is a nonzero element whose square is zero and whose scalar image would be a nonzero complex number whose square is zero. For the anticommuting form, a unital homomorphism into the scalars sends units to nonzero

scalars, and applying it to the anticommutation relation forces the product of the two image scalars to vanish. The interface statements compose these arguments with the declared restriction into the scalar or commutative region. The inhabitant is finite linear algebra over one declared regulator with identity repair maps; the general matrix obstruction, the interface compositions, the interface laws, both noncommutativity witnesses, the mutual noncontainment, and the corner-character retractions are checked in Lean [7]. □

The declared finite C^*-algebra/retraction interface is therefore compatible with noncommutative regional coexistence through the displayed character-block witness. The obstruction excludes the full-matrix and anticommuting presentations stated in the theorem; it does not classify every possible coexistence presentation. The declared overlap algebra is a proper subalgebra of the set intersection of the two block algebras, and genuine coverage semantics or regional-factor receipts for a source-produced net are not supplied.

Dynamics on the private block carries a matching rigidity statement.

Theorem 2.2 (Finite Stone converse on one full private block)**.** *Let α_t, $t \in \mathbb{R}$, be star automorphisms of one full finite matrix block $M_n(\mathbb{C})$ with $\alpha_0 = \mathrm{id}$, $\alpha_{s+t} = \alpha_s \circ \alpha_t$, and every orbit map $t \mapsto \alpha_t(x)$ continuous. Then there is a self-adjoint $H \in M_n(\mathbb{C})$ with*

$$\alpha_t(x) = e^{-itH}\, x\, e^{itH} \qquad \text{for all real } t \text{ and all } x \in M_n(\mathbb{C});$$

the conjugating family $U_t = e^{-itH}$ is unitary with $U_0 = 1$ and $U_{s+t} = U_s U_t$, and H is unique up to one additive real scalar multiple of the identity.

Proof. Continuity upgrades to differentiability at parameter zero by interval averaging: the group acts on the matrix-unit basis as a norm-continuous matrix family whose short-interval average is invertible, and the fundamental theorem of calculus recovers the family from its primitive. The derivative at zero is a star-compatible Leibniz endomorphism. Every complex-linear Leibniz endomorphism of the block is an inner derivation, with an explicit matrix-unit witness proved from scratch; star compatibility symmetrizes the witness into one self-adjoint generator; two generators with one commutator action differ by a real scalar multiple of the identity; and the group differential equation identifies the flow with conjugation by the coherent unitary family. Every step is checked in Lean [7]. □

On the finite private block, a pointwise-continuous symmetry flow of the observables therefore has no choice except Hamiltonian form; the additive energy offset is the only freedom. The theorem extends to the general finite private algebra: every unital star subalgebra of a finite complex matrix algebra, containing the ambient identity, is star-isomorphic to a finite direct sum of full matrix blocks, and on that block model a pointwise-continuous star-automorphism group fixes every block projection and is blockwise conjugation by the unitary family of one time-independent self-adjoint Hamiltonian per block. Block fixing is derived rather than assumed; a single automorphism may swap isomorphic blocks, and the swap is realized on the square two-block algebra. The parameter is a real number; no physical clock, source-selected Hamiltonian, or physical time identification is claimed.

The measurement and dynamics surfaces share one object. A composed frame-duality theorem (`schroedinger_frame_duality`) consumes the registered record representation, operational effect additivity, and one supplied pointwise-continuous flow through a single typed bundle and proves that the unique Busch–Gleason state of the Heisenberg-shifted registered frame is exactly the propagator conjugation of the unique state, at every flow parameter: the Schrödinger picture of the Born frame is forced by the Heisenberg action of the rigid generator. The flow stays a supplied

premise and its parameter is not physical time; what the theorem removes is any freedom to represent the measurement statistics and the dynamics on different state objects.

3 Records and Quantum Identities

3.1 Finite event algebra

The span of a projective record partition is a commutative star subalgebra exactly star-isomorphic to complex functions on its nonzero projector labels. A common linear isometry can copy two distinct sharp states from one normalized blank only when they are orthogonal; the full mixed-state no-broadcasting implication enters through an explicit adapter. Partition pinching and partition averaging have explicit normalized Kraus families and preserve trace, and a separate module proves the averaging, pinching and nonnegative-parameter relaxation channels completely positive and trace preserving. The two-scale public relaxation and Poissonized repair families obey exact semigroup laws, and on bounded endomorphisms of a real Banach space the latter is exactly $\exp[t\gamma(E-I)] = E + e^{-\gamma t}(I-E)$; this operator exponential supplies no source rate or physical clock.

Positive unital maps of the active-record function algebra are exactly real row-stochastic kernels. Every star automorphism of the public function algebra is uniquely pullback by a label permutation, so every pointwise-continuous real-parameter group of public star automorphisms is trivial. On one finite full private matrix block, every star automorphism is unitarily inner, and a fixed self-adjoint Hamiltonian gives a unitary real-parameter flow obeying the von Neumann equation; the converse is Theorem 2.2. For nonselective projective pinching, $S(\mathcal{E}\rho) - S(\rho) = D(\rho\|\mathcal{E}\rho) \geq 0$; this analytic matrix identity is separate from a normalized selective Lüders branch. Arbitrary partition pinching is exactly the uniform random-unitary average over all 2^k independently signed block reflections. A complementary support boundary is exact: the totalized matrix logarithm sends every projection to zero, so its raw relative-entropy formula assigns zero to an orthogonal pure-state pair whose support inclusion fails. The projective partition and input state are finite source data, and the public automorphism classification is complete for the exact finite active-record function algebra and its matrix realization.

3.2 The Born weight: effect-valuation theorem and frame boundary

Theorem 3.1 (Finite effect-valuation representation)**.** *Let $\mathcal{A}$ be a finite matrix algebra. Every valuation on the effects of $\mathcal{A}$ that is nonnegative, normalized and additive on coexisting effects equals $E \mapsto \mathrm{Tr}(\rho E)$ for a unique density operator ρ. This holds in every finite dimension including two, with no continuity axiom.*

The theorem is checked in Lean [7]. Its premise, noncontextual additivity on all coexisting effects, is the load-bearing clause, and the finite receipts locate it rather than derive it. The twelve declared central port atoms form one classical context with an eleven-dimensional weight simplex. A distinct qubit spinor adapter has six disjoint antipodal binary contexts whose additive weights have affine dimension six; the trace-one Hermitian/Born slice is three-dimensional and obeys three exact golden-ratio relations. Representation there is unique when it exists, but exact weights show that neither existence nor positivity follows. On the full celestial or Bloch sphere, the continuous weight $F(n) = (1 + n_z^3)/2$ is normalized on antipodes yet non-affine. Sharp binary webs admit the cube valuation as a non-Born extension, and one unsharp three-effect context excludes that particular extension. The same finite battery admits the planar cubic response $F_y(n) = (1 + n_y^3)/2$,

which agrees with the maximally mixed Born weight on every displayed axis while remaining non-affine away from them. The projector webs alone therefore underdetermine the weight, and one finite unsharp battery does not force a global valuation.

The real S_3 algebraic web attached to source-realized gauge labels is blind to the Pauli-Y direction: two distinct complex states give equal weights on every declared projector candidate. The missing algebraic direction is exact. If P is the declared diagonal projector and Q its noncommuting image under the declared two-dimensional representation of a source-realized gauge label, then

$$\frac{I}{2} - \frac{2\sqrt{3}}{3} i(QP - PQ) = \rho_{Y+}.$$

Together with P and Q, this phase lift separates every fixed-trace 2×2 matrix. Conversely, every effect generated by real linear coarse graining, complements and real Kraus pullbacks remains real symmetric and gives the same weight to ρ_{Y+} and ρ_{Y-}. Entrywise conjugation fixes the complexified real web pointwise and exchanges ρ_{Y+} with ρ_{Y-}, so the ambiguity between the two candidate completions is one conjugation orbit, and any conjugation-fixed frame conflates each state with its conjugate. The phase lift lies in the complex operator algebra generated by the source-attached pair; it is a declared effect, not an operation or instrument.

A post-hoc statistic on one bounded run, executed under an unrelated locally hash-pinned common-reference contract, is reversal-odd on raw counts: a designated pair counts 1343 against 0, and a designated three-cycle has count products 1239691068 against 0, machine-checked as the unique maximizing orbits of the post-hoc designation rule. Transposing the counted order flips the selection bit, and a synthetic symmetric count table fails the strict-inequality applicability condition. Under a declared transport convention the selected completion coincides with the phase lift, either candidate completes state tomography, and Theorem 3.1 pins the represented state by the three completed-frame weights. The opposite pairing is equally admissible, neither the statistic nor its designation rule was preregistered, and the counted orientation carries no arrow-of-time claim.

On the committed static effect and count fixture this boundary is exact. Every committed effect has trace one, so a sum of two committed effects is an effect precisely when it resolves the identity, and any assignment sending the sure effect to one and normalized in each context is additive on every coexistent sum formed within the committed set. A cubic deformation of the run Born functional, shaped to vanish on the committed set, reproduces every generated expected frequency, stays in $[0, 1]$ on all effects, obeys the complement rule and is additive on every committed coexistent sum, yet it fails additivity at an explicit effect pair with values $35/64$ against $143/256$ and is the Born functional of no state. Conversely, full cross-context additivity together with the static fit values forces the Born functional of the declared matrix uniquely. Between the committed clauses and full cross-context additivity lies an injective affine family through the run Born member with additivity residue exactly $-(3/256)\,t$ at the witness pair. On the convex interval $0 \le t \le 1$ every member is probability-valued on every effect; outside that interval only the algebraic identities are asserted. Full additivity selects the Born endpoint from the probability-valued continuum.

A declared Lüders instrument on the committed effect table supplies completely positive outcome maps, a trace-preserving summed channel per context and induced effects equal to the table; the table does not determine the instrument. The placeholder binding schema splits static compatibility fields from the evidence a source-bound run would have to supply. Its determined matrix and declared expected-frequency table are inhabited by every compatible declared instrument; they are not a source-produced preparation, public outcome or readback. The record's natural-number digest and inhabited proposition fields are freely stipulable, so the schema authenticates no producer or custody artifact. Positive closure of the Born attachment therefore requires a source-realized complex-tomographically complete effect and instrument web, an operational coarse-graining or

equivalence theorem deriving noncontextual additivity, and only then preregistered frequencies as validation; frequencies alone do not prove the universal valuation law.

3.3 Record algebra, measurement, Bell correlations and checkpoints

The fixed-cutoff record, measurement, Bell and checkpoint packages are proved in the screen-microphysics paper [8] and the consensus paper [6]. They are stated here in the form the observer layer uses.

Public normal forms. On its declared fixed-cutoff branch, the consensus construction gives a finite constraint code whose codewords are the globally consistent states $C = \Phi^{-1}(0)$; it is not automatically a quantum error-correcting or topological code, and its graph min-cut does not determine code distance. For the declared accepted repair law, the local-fit contract makes Φ a Lyapunov functional, the union-collar gluing package supplies the local diamond on the physical quotient, and that confluence condition together with repair completeness yields a unique schedule-independent normal form from a fixed initial quotient state. On the abelian branch, global consistency holds exactly when cycle holonomy vanishes; on the noncentral branch the crossed-module gluing orbit $q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$ labels the fixed-cutoff gluing orbit. The repair law descends to the overlap-invariant quotient, and the induced terminal state on the declared physical observable algebras is unique there even when microscopic representatives differ by gauge or sector relabelings.

Record algebra and the Born–Lüders package. For each completed compare/write/verify slice of the regulated microphysics, the declared pointer and overlap-sector projectors generate a finite commutative central record algebra $\mathcal{Z}_{\mathrm{rec}}$. Given the supplied density-state valuation and declared Lüders instrument, every observer-accessible event E in that algebra has probability $\mathbb{P}(E) = \mathrm{Tr}(\rho P_E)$, and the selected conditioning on E gives

$$\rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

The effect projectors do not determine this instrument; same-effect non-Lüders instruments remain possible. If a practical readout uses projectors Q_a with commuting central reference projectors $\widehat{Q}_a$ on the same slots and $\delta_{\mathrm{rec}} = \max_a \|Q_a - \widehat{Q}_a\|$, then $\|[Q_a, Q_b]\| \le 4\,\delta_{\mathrm{rec}}$, and any accessible-state perturbation $\|\widetilde{\rho} - \rho\|_1 \le \varepsilon$ changes each declared elementary record-event probability by at most $\varepsilon + \delta_{\mathrm{rec}}$. Records carried by central projectors are shareable without violating no-cloning for generic quantum states.

Native preparation for a complete quantum readout. The native total-two carrier has dimension 78, with a 66-dimensional binary subspace embedded by J. A fixed sweep of its thirty seams removes collision support. Eighteen further sweeps contract the diagonal deviation and off-diagonal coherence. For every initial density ρ_{SR}, including entanglement with an untouched finite reference R, the 570-attempt preparation satisfies

$$\frac{1}{2}\left\| \Phi(\rho_{SR}) - J\frac{I_{66}}{66}J^* \otimes \rho_R \right\|_1 \le \varepsilon < 3.7187 \times 10^{-8}.$$

The bound uses exact rational sweep powers and a uniform coherence contraction over the admitted native record laws; it does not claim exact equilibration at finite time.

Compose this preparation with the complete capped phase/work instrument. At cap 64 its 8,195 suffix records retain failed heralds, both herald signs, phase bits, all work records and unmeasured

timeouts. Contractivity adds ε once to the complete output comparison, independently of the cap, record count or useful-herald probability. The warm-up tape is ignored without selecting trials, and subsequent controls cannot consult it. The original reference marginal is preserved. Conditioning on a suffix herald requires a separate error bound.

The compiled coherent mixer has an additional full-carrier half trace-norm allowance δ, so its complete comparison to the exact stationary ideal reference is bounded by $\varepsilon+\delta$. Since the numerical binary mixer need not preserve trace exactly, individual event weights use $\varepsilon + 2\delta$. The proofs and numerical-reference allowances are in `model/prepared_quantum_readout` and `model/clocked_quantum_readout`. The result supplies finite native preparation in place of exact $I_{66}/66$ and initial binary support. The total-two sector, quantum instrument semantics, coherent service and work action, preparation and clock retain their stated premises. This is neither a Born-rule derivation from scalar consensus nor a physical thermal equation of state.

Bell correlations. On the declared fixed-cutoff observable algebra, fix commuting left and right wing subalgebras, central binary setting registers, binary projective readouts on each wing and a source state on the joint wing algebra. Then

$$p(a,b \mid x,y) = \mathrm{Tr}\big(\rho_{LR}\, P^{(L)}_{a|x} P^{(R)}_{b|y}\big), \qquad E(x,y) = \mathrm{Tr}\big(\rho_{LR}\, A_x B_y\big),$$

the local marginals are independent of the remote setting, and $\big|E(0,0)+E(0,1)+E(1,0)-E(1,1)\big| \le 2\sqrt{2}$. An explicit two-qubit branch with the stated Pauli readouts saturates $2\sqrt{2}$; Bell-pair preparation and two-qubit factor structure are branch conditions. Reachable states on the committed pair carriers are separable across the declared slot split and obey the classical bound 2 against every slot-local unit-interval readout, jointly diagonal or not; a diagonality-preserving map cannot reach the declared Bell-state witness, and a product-diagonal state paired with cross-commuting settings outside both slots reaches $2\sqrt{2}$. Slot membership relative to the split therefore carries the classical bound.

Collar factorization, checkpoints and restoration. For a collar tripartition A-B-D, the exact Markov normal form is used only when $I(A:D|B) = 0$ holds literally, or along a controlled fixed-collar family whose exact-Markov replacement modulus tends to zero. Then

$$\rho_{ABD} = \bigoplus_{\alpha} p_\alpha\, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D},$$

with classical sector label α in the center. A checkpoint is the tuple $\mathcal{C} = (R, \alpha, \rho^{(\alpha)}_{\mathrm{int}})$, and for a compatible target environment state the restored state is $\rho^{(\alpha)}_{\mathrm{int}} \otimes \sigma^{(\alpha)}_{\mathrm{env}}$. For approximate Markov collars, with ε the conditional mutual information in nats, a Fawzi–Renner recovery channel gives

$$\big\|\rho_{ABD} - (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})\big\|_1 \le 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

The fixed-cutoff checkpoint theorem is stronger than this recoverability estimate: for an observer patch whose checkpoint consists of the observer-facing record algebra, the accessible local state, the future update schedule and the visible overlap interface data, exact restoration reproduces the full future law of observer-accessible events, and an ε-accurate restoration changes that law by at most ε in total variation [8]. The package is same-interface restoration; substrate transfer, continuation across redesigned environments and uniqueness of an observer-supporting sector are outside it.

3.4 Finite regional observer nets

A proof-carrying enrichment of one finite consensus tower packages a finite region order, isotony, exact commutation on declared-disjoint regions, covariant refinement, compatible regional expectations and idempotent local repair. A declared repair that fixes remote observables gives exact Heisenberg nondisturbance for every matrix state. Compatible local sections have a unique glue only on nonempty finite subregion families carrying the explicit restriction and uniqueness receipts; the interface name `FiniteCover` includes no joint-coverage law. A separately declared basis split and Kraus completeness give a generic reindexed marginal identity, and no theorem identifies the regional algebras with its matrix factors. Contravariant restriction maps and their retraction of isotony are strong extra premises, not consequences of an ordinary isotonic net.
A rich-fibre packet gives a stronger conditional construction on one preregistered fresh-seed bounded run: four target-blind greedy-disjoint observers realize split-fibre counts $3, 3, 4, 4$ on pairwise-disjoint twenty-node windows, and the mirrored net carries a noncommutative block algebra with an embedded two-by-two designated-fibre matrix corner at every observer region, kernel-backed window disjointness, elementwise commutation across distinct windows, the character-and-block restriction laws, the computed four-window coverage of the top algebra, and a drop-one control showing that every window is load-bearing. The enrichment rule, region lattice, restriction system, base-point state and identity repair are declared readings, and the matrix corner is not a regional tensor factor.

For each of four retained observers, counted transition operators together with diagonal field projectors generate the full matrix algebra on the realized alphabet. The fully support-disjoint retained pair 86/247 defines a positive rational correlated state with exact counted marginals, recovered by a conditional-expectation diamond whose regional pair erases the correlation, so coverage cannot imply unique reconstruction. A typed linear join embeds the 182-dimensional 86/88 and 169-dimensional 86/247 diagonal layers into a 338-dimensional pushout over their shared 13-dimensional observer-86 record space; the pushout is a linear record construction whose quotient has no induced multiplication. On the same three-observer label carrier, identity-spectator embeddings of the full pair matrix algebras intersect in exactly the full M_{13} hinge and generate M_{2366} (Section 3.5). A separately declared slot exchange gives an alternating regional family whose evolved admitted data generate the next regional algebra. The path-read permutations and the slot exchange are supplied adapters; physical regions, quantum preparations and outcomes, a physical clock and a Cauchy-surface interpretation are outside these finite statements.

Theorem 2.1 bounds every such net: no region containing a unital full matrix algebra of dimension at least two sits above a scalar region, while the character-block witness realizes noncommutative coexistence over a scalar overlap.

3.5 A noncommutative join of the retained observer carriers

The retained observers 86, 88 and 247 have realized label spaces of dimensions 13, 14 and 13. Their 32 aligned source rows define one triple path $q_t = (a_t, b_t, c_t)$, $0 \le t \le 31$, with the two pair paths as its coordinate projections. Use the declared tensor reading $H = (\mathbb{C}^{13} \otimes \mathbb{C}^{14}) \otimes \mathbb{C}^{13}$, and write $M_n = \mathrm{End}(\mathbb{C}^n)$. The operator join uses this 2366-dimensional carrier; it assigns no multiplication to the 338-dimensional linear pushout of diagonal records.

Theorem 3.2 (Full operator join on the common carrier)**.** *The maps $\iota_{88} : M_{182} \to M_{2366}$ and*

$\iota_{247} : M_{169} \to M_{2366}$, *defined by*

$$\iota_{88}(X) = X \otimes I_{13}, \tag{1}$$

$$\iota_{247}(Y)_{(a,b,c),(a',b',c')} = Y_{(a,c),(a',c')}\delta_{bb'}, \tag{2}$$

are injective unital star homomorphisms. Their images intersect in exactly $M_{13} \otimes I_{14} \otimes I_{13}$ *and generate all of* M_{2366}*. The exclusive observer-88 and observer-247 factors commute. The two overlapping pair algebras do not commute.*

Proof. Identity-spectator inclusion and coordinate reindexing preserve products, adjoints and the identity. Evaluating a fixed spectator diagonal entry proves injectivity. If $\iota_{88}(X) = \iota_{247}(Y)$, evaluation at $c = c' = 0$ gives

$$X_{(a,b),(a',b')} = Z_{a,a'}\delta_{bb'}, \qquad Z_{a,a'} = Y_{(a,0),(a',0)}.$$

Thus $X = Z \otimes I_{14}$, and injectivity forces $Y = Z \otimes I_{13}$. Conversely these two matrices have the same triple image. This proves the full M_{13} intersection, whose complex vector dimension is 169.

For any triple indices $q = (a, b, c)$, $r = (a', b', c')$, and any hinge label s, the pair matrix units obey

$$\iota_{88}(E_{(a,b),(s,b')})\, \iota_{247}(E_{(s,c),(a',c')}) = E_{q,r}.$$

Every triple matrix unit therefore belongs to the generated algebra. Operators on the exclusive factors commute by the tensor-product rule. The common hinge contains E_{01} and E_{10}, whose two products are the distinct E_{00} and E_{11}, so the pair algebras cannot commute elementwise. □

Counted states and observable retractions. The empirical state and its pair marginals are

$$\rho_{ABC} = \frac{1}{32}\sum_{t=0}^{31} |q_t\rangle\langle q_t|, \qquad \rho_{AB} = \mathrm{Tr}_C\, \rho_{ABC}, \qquad \rho_{AC} = \mathrm{Tr}_B\, \rho_{ABC}.$$

For arbitrary pair observables, these give

$$\mathrm{Tr}(\rho_{ABC}\iota_{88}(X)) = \mathrm{Tr}(\rho_{AB}X), \qquad \mathrm{Tr}(\rho_{ABC}\iota_{247}(Y)) = \mathrm{Tr}(\rho_{AC}Y).$$

Observable retractions instead use normalized partial traces:

$$\frac{1}{13}\,\mathrm{Tr}_C\, \iota_{88}(X) = X, \qquad \frac{1}{14}\,\mathrm{Tr}_B\, \iota_{247}(Y) = Y.$$

These reverse maps are linear, not multiplicative. The density-state marginals carry no spectator normalization. The empirical states are distinct from the independently supplied uniform tower states; no state-preserving tower/refinement morphism follows from these equalities.

Source generators and finite transport. Each observer's counted transition operator and diagonal field projectors generate the full matrix algebra on its realized alphabet. Their pair lifts, followed by ι_{88} and ι_{247}, therefore generate the full triple algebra. This is a family of operators constructed from source data, not a history of source-executed quantum gates. A singleton path projector does not replace this generating family.

For each $0 \le t < 31$, transpose the two adjacent path labels separately in each factor, and let U_t be the tensor product of the resulting permutation matrices. The triple conjugation α_t and its pair conjugations satisfy

$$\alpha_t\iota_{88} = \iota_{88}\alpha_t^{88}, \qquad \alpha_t\iota_{247} = \iota_{247}\alpha_t^{247}.$$

These identities hold for arbitrary matrices, including off-diagonal observables. The evolved source generators generate the full joint algebra because α_t is an automorphism. Set $P_0 = I$, $P_t = U_{t-1} \cdots U_0$, and $U_{s,t} = P_t P_s^{-1}$. Then $U_{r,t} U_{s,r} = U_{s,t}$ for $s \le r \le t$, with exact pair projections and path transport. The final-to-initial edge is absent from all three transition-count tables and is excluded. The empirical state is not invariant under these row-dependent permutations.

The general operator identities, counted-state readouts and evolved source generation are kernel checked in `TripleCarrierOperatorJoin`. Independent source and permutation replay in the public repository [7] checks all 31 adjacent transitions and 496 nonempty ordered row intervals; its sparse matrix-unit controls are finite examples, separate from the general proofs.

The label extraction is post hoc; the tensor assembly and path-read transpositions are supplied adapters. They provide no preregistered physical validation. Observer 86/88 source supports overlap, so their tensor slots do not prove physical support disjointness. This construction supplies no universal algebraic pushout, physical region assignment, quantum preparation or outcome record, physical clock, source-selected Cauchy geometry or interacting continuum limit. It does not identify this carrier with the separate 64-mode scalar instrument.

4 Common-World Joins on the Twelve-Port Carrier

This section places one observer's mass, clock, worldline and a declared Maxwell evolution on one typed record of the twelve-port carrier. Every join below is a declared regression surface: it shows which identities the shared record forces and which choices remain free, and none is a source-selected physical action. Six receipts of the formal stack share one typed record. A common-world architecture record carries one declared positive mass, one state frame, one flow frame, one projective partition with state, one declared Maxwell evolution bundle on the twelve-port local carrier, and one declared electroweak premise bundle whose base instantiates the Standard-Model structure premises. For every inhabitant, the kinematic mass-shell, internal-clock, free-evolution, and causal-order receipts hold of the same mass, frame, flow frame, partition, and state on one Hermitian Lorentz module, with the shared fields identified definitionally, and four join theorems place the four-momentum, the clock worldline, and every nonnegative flow displacement inside the cone order that carries the finite causal nets, with the four-momentum future-causal from the causal net's bottom vertex. Those nets and their Lorentz-module cone order are declared kinematic data, not the source-derived informational read-from order. On the screen side, the temporal Maxwell receipt and the electroweak-breaking receipt hold of the same record, and the pinned port-dual measure of the record's structure premises evaluates the Maxwell Gauss load on the one twelve-port index type, with every weight equal to $1/12$. An explicit instance inhabits the record (`CommonWorldKinematicsWitness`). On the base record the two islands are joined only by conjunction: no typed dictionary sends ports, seams, or faces to points or cones of the module, the identification of the Maxwell step index with the kinematic proper time is a declared dictionary carried by the instrumented extension below, and no common variational principle spans both carriers; clock calibration and carrier identification are not supplied. Gravitation, thermodynamics, matter dynamics, quantization, and observer readout appear in no field of the record, so the record does not establish the common-world target.

One typed dictionary joins the two islands beyond conjunction. The rank-three source Gram quotient of the six-axis port adapter, read from the registered twelve-port incidence table, maps onto the rest fiber of the record's declared frame by a linear equivalence composed from committed isometries, and the pullback of the rest metric through that dictionary is exactly the screen quotient Gram at every declared frame, with the standard-frame case equal to the committed can-

didate readback and with every integer record control carried to the committed screen chain up to one committed normalization radius. An extended record carries the dictionary and the metric clause as fields; an explicit inhabitant extends the committed instance, and a composed receipt proves the eight island packets, the bridge clause with its oriented-chart commuting square, and the transported inverse-square strength read on the screen Gram, of one record simultaneously (`CommonWorldIslandBridge`). Dropping the metric clause readmits a mismatched pair: the same base witness extended by the dictionary scaled by two is well typed and violates the clause at an exhibited unit class. The dictionary is a candidate local readout between two mathematical objects and identifies nothing physical; the shared clock, the common action, common dynamics, the map from ports, seams, and faces to the module, and the physical spacetime attachment are not supplied.

Four further typed joins extend the bridged record. A step-scaled Maxwell bundle sits beside the declared unit-step bundle on the same twelve-port carrier and shares its initial seam data and seam current; the unit-step law is the unit instance of the scaled law, every certified Courant constant is at least the committed eigenvalue 5, so every certified step lies below one, and the receipt carries the explicit unbounded unit-step mode beside the uniform bound on the scaled fields. The phase instrument joins on the two-dimensional record algebra: its diagonal-context channel is exactly the committed pinching of the diagonal partition, which refines the record's one-block partition as every partition does, and the swap-twisted instrument violates that identity at the run state. A declared positive step duration δ identifies evolution step n with the proper-time parameter $n\delta$ on the record's clock worldline: the clock at step n is $e^{-imn\delta}$, it repeats after N steps exactly when $mN\delta \in 2\pi\mathbb{Z}$, successive step events are causally ordered, and a zero or negative step collapses the steps or breaks the causal clause. A repair law whose reference is the diagonal of the instrument's preparation carries a horizon record, and the Einstein-branch register at that law states the first-law clause on the simplex tangent space (`CommonWorldInstrumentJoin`). A formal same-index product readout packages the two islands: a single function of the step index produces the pair of the scaled staggered-form value of the record's certified bundle and the clock worldline event at proper time $n\delta$, with the index recovered from the joined object, successive joined events causally ordered, the record's step below the sharp certificate, and, for zero seam current, the joined value constant along every causally ordered index pair while the electric and magnetic energies stay within the committed multiples of the joined value at index zero. These identities are inherited independently from the event and field projections; no theorem relates the field value to the event beyond the shared index. On the committed inhabitant the joined value is constant while the clock does not repeat with period one step, and the product fixes no clock calibration (`CommonWorldMaxwellClockJoin`). A chosen quadratic clock functional can be added formally to the Maxwell functional: conditional on its declared endpoints, the worldline is the unique stationary path in the committed Lorentz form, whose exact expansion carries the action of the variation as the entire quadratic remainder, and the direct-sum action, the committed scaled Maxwell window action plus the worldline action on the shared window, is stationary in its Maxwell sector exactly at the scaled Ampère and Gauss clauses and in its worldline sector exactly at the record's clock line, with the committed configuration stationary in both sectors. The sum is proved a direct sum, partial variations independent in both directions, and because the form is Lorentzian a displaced spatial history strictly beats the stationary worldline, so stationarity is proved without any minimality claim (`CommonWorldJointAction`). This zero-coupling sum is a regression surface, not a source-selected physical common action; its displayed relative sector coefficient is likewise declared and cannot be fixed by the decoupled stationarity equations. The step dictionary is declared and identifies no physical time. For the carrier map itself a candidate exists: twelve exact golden-ratio-ring rays embed the ports into the spatial slice of the committed Lorentz module, reproduce

the committed Gram table up to one squared-length normalization, respect the seam, face, and antipode incidence in both directions, carry the full sixty-element committed action through two exact rotation matrices and a sixty-word generation certificate, act compatibly with positive-factor barycentric refinement on nondegenerate mesh points, and land spacelike for the committed determinant form; the canonical assignment and its negation both satisfy every clause, so the selection is declared rather than forced, and the embedding carries no scale, causal content, or observer readout (`ScreenCarrierMapCandidate`). The candidate is dynamics compatible: a constrained transport structure, whose two generator inhabitants are kernel-verified row by row against the committed seam and face tables, intertwines the scaled Ampère update, the Gauss constraint with pulled-back charge, the staggered energy form, and the committed gauge transformation exactly and uniformly in the step, so no calibration is selected; through the committed word certificate every row of the sixty-element action is the port component of such a transport, the generators act geometrically through the two exact rotation matrices, and transport commutes with barycentric refinement along both routes. A declared interaction candidate couples the worldline increment to the seam electric field through the embedded seam directions: it is gauge invariant at every relative normalization, its zero value recovers the decoupled direct sum as the regression limit, and an explicit pulse configuration separates normalizations, so the coupling constant is declared rather than forced (`CarrierDynamicsCompatibility`). At a declared nonzero step h, varying the coupled action in both sectors shows what the candidate couples: the field equations are the committed Ampère and Gauss clauses at augmented sources, and the induced sources are polarization-type, a step difference of the worldline seam current and a neutral bound load, so the candidate carries no monopole charge and its force on the worldline is the coupling times the step difference of the embedded field, with the exact worldline equation $2m\,\Delta^2 x = \kappa\,\Delta E$. One declared identification of the induced Gauss-law load at a port with a declared value q fixes the normalization uniquely, $\kappa = -qh/(12 - 4\varphi)$ on a seam crossing, with the constant computed exactly from the golden-ratio seam data; the identification is step-local and increment-proportional, only κ/h is fixed, and the minimal-coupling shape pairing the potential itself with the increment is the live monopole alternative (`ChargeFixedInteraction`). The dynamical content of the mass–energy identity is located in the same setting as a conditional theorem: under a declared composite shape in which the rest parameter plus the modular energy of the cap first law multiplies the committed worldline functional, stationarity is exactly $(m + E)\,\Delta^2 x$ equal to the impulse of a declared potential, so internal energy enters the inertial coefficient with slope one, while a second declared shape entering the same ledger additively has coefficient m, agrees with the first at zero energy and on the committed worldline, and is separated from it by an explicit parabola history; the slope with which internal energy enters inertia is selected by the shape, both shapes are declared enrichments by the realized-history Legendre non-identifiability, and nothing derives the identity (`InternalEnergyInertia`).

The monopole route named there is built as a declared point charge hopping along the ports of the thirty-seam complex, minimally coupled to the seam and port potentials with the source weights of the committed window action. At a declared nonzero step h, its induced load has total q at every step and its induced current is $-(q/h)$ on the hopped seam, oriented so that the committed continuity equation holds identically; the coupled action equals the window action at the augmented sources with no endpoint term, so the field equations are the committed scaled Ampère update and Gauss constraint at the augmented sources. For a general source pair the gauge change of the source pairing is the gauge function paired against the continuity residual plus an endpoint term, and invariance under endpoint-vanishing gauge functions is equivalent to continuity on the interior steps. Along the Ampère evolution the staggered energy moves by $(q/2)(E_n(e) + E_{n+1}(e))$ on a forward hop across e and is constant at rest; no energy of the charge and no equation of motion for the path are defined. At the endpoints of a crossed seam the E-paired induced load is $-(\kappa/h)(12 - 4\varphi)$

times the step difference of the unit hopping load, and equals the hopping step difference exactly at $\kappa = -qh/(12-4\varphi)$; at the exhibited seam a port away from the endpoints carries a nonzero E-paired load while the hopping difference vanishes, so the polarization is the step difference of the monopole at the seam endpoints only. At nonzero h the charge is unconstrained by the field sector, and the path, charge, nonzero step, and coupling shape are declared (`PortChargeMinimalCoupling`).

The monopole route joins the clock action through a declared class of worldlines. A seam-step worldline is a declared start port and a declared step sequence, each step a rest or a signed seam of the carrier map candidate; it generates a Lorentz-module path with time coordinate τn for a declared unit τ and spatial coordinate the cumulative sum of the signed seam vectors, and a hopping port path. The seam vector is the ray difference of its endpoints in $\mathbb{Z}[\varphi]^3$, and at every step the ray of the occupied port minus the ray of the start equals the spatial coordinate exactly, so the injective ray map reads the port off the path. At every crossing step and a declared nonzero field step h, the E-paired induced load at the two endpoints of the stepped seam equals $-(\kappa/h)(12-4\varphi)$ times the step difference of the unit hopping load, and equals the hopping step difference of charge q at $\kappa = -qh/(12-4\varphi)$; the worldline seam current is the $\mathbb{Z}[\varphi]$ pairing of each seam vector with the stepped vector, ± 4 at the stepped seam and at its antipodal seam (the two carry one vector) and nonzero at a neighbouring seam, and the hopping current is $-(q/h)$ times its projection onto the stepped seam divided by four. At nonzero h, the transported action, the monopole coupled action plus the clock action of the generated path, has field-sector stationarity equal to the committed Ampère and Gauss equations at the transported sources; the clock action is $\sum(\tau^2-4)$ over crossings plus $\sum \tau^2$ over rests, timelike exactly when $\tau^2 > 4$. Distinct units give distinct paths with one hopping path and one source family, and at nonzero h the unit is unconstrained by the field sector (`WorldlineHopTransport`).

The transported charge has a discrete balance. Varying the transported action over the declared class of closed two-step variations, the replacement of one step pair by an admissible pair with the same endpoint, gives an exact action difference: the clock part is the original step norms minus the varied ones (rest 0, crossing 4, the unit cancels), the interaction part is q times the potential pairing $h\varphi_n(u) - \langle \mathbf{1}_{u\to v}, A_{n+1}\rangle$ summed along the alternative route minus the original one, and the window action cancels. Exchanging a forward crossing of seam e with an adjacent rest moves the action by $q\,h\,E_{j+1}(e)$ at zero clock cost, so, for $q \neq 0$ and $h \neq 0$, the exchange is stationary exactly when the scaled electric seam field vanishes at the delayed step; replacing two rests by a round trip across e moves it by $-8 - q\,h\,E_{j+1}(e)$, so at zero field rest is neither stationary nor a local minimum in this class, the port form of stationarity without minimality. For static potentials the interaction difference is q times the port-potential difference at the intermediate port plus q times the discrete circulation of A around the four-leg loop, and a round trip on one seam has zero circulation. The class, the potentials, and the shapes are declared (`TransportedChargeForceLaw`).

The closed carrier also fixes the neutrality condition rather than hiding it in the field solver. A hopping load and current obey the discrete continuity equation identically; the sourced scaled Ampère and Gauss equations admit a history exactly when the initial load is neutral and continuity holds, so one isolated hopping charge is compatible only at zero charge. The viable neutral branch is explicit. Put charges q and $-q$ on two declared seam-step worldlines and form one action from the source-free Maxwell window, both minimal couplings, and both clock actions. Its field variations are equivalent, for $h \neq 0$ and endpoint-fixed seam-potential variations, to the interior Ampère equations at $m < N$ and Gauss equations at $n < N+1$ for the pair's own sources. The action is invariant under gauge functions vanishing at the window endpoints. Replacing either path over the same closed two-step class, with both fields and the other path fixed, gives an exact off-shell clock-plus-interaction action difference, not by itself an equation of motion. At $q = 1$ and $h = 1/2$, crossings of seams 0 and 29 recover an explicit nonzero-current, Coulomb-started field-stationary inhabitant.

It fails the declared closed-two-step exchange-stationarity condition in both path slots: for every $N \geq 1$, delaying either the positive seam-0 crossing or the negative seam-29 crossing while fixing the fields and the other path changes the same action by $5/12$ for every pair of clock-unit values within the fixed quadratic clock-action form. The two results are charge conjugates; applying both canonical exchanges simultaneously changes the action by $5/6$. These are only the displayed discrete exchanges, not continuum first-variation claims or a no-go for other paths, fields, or action shapes. The counterexample is specific to that Coulomb start. On the window $N = 1$, a second exact rational temporal-gauge history at the same $q = 1$, $h = 1/2$, and the same crossing pair satisfies the sourced Ampère and Gauss equations. The four possible intermediate ports for each closed two-step path replacement are exhausted exactly. Endpoint routes have zero action difference, and the two triangular detours have clock difference -4 cancelled by interaction difference $+4$. Thus one finite action is stationary in both field slots and unchanged under every in-window closed two-step replacement in either path slot. The seam currents are -2 and $+2$ on seams 0 and 29, and the initial loads are $+1$ and -1 at the corresponding ports. At the separately declared clock unit $\tau = 3$, both crossings are timelike in the carrier's Lorentz-module metric. The charge, paths, step, rational history, clock unit, coupling shape, and variation class remain declared; the theorem establishes existence without uniqueness or source selection. It proves no minimum, stability, or dynamic selection. No laboratory-time calibration or continuum limit is claimed (`SeamChargeContinuity`, `NeutralPairCoupledAction`, `NeutralPairJointStationaryWitness`).

Inside the rest-diluted class the law reads as a position law. A general one-step replacement with shifted later ports has the exact difference of the one-step clock term plus q times the pairing difference at the step plus q times the sum of the later pairing differences at the shifted ports, reducing to the one-step term under declared shift invariance and to the closed two-step difference when the route rejoins at the second step. Within a block of k rests and one crossing, moving the crossing by one position changes the action by $q\,h\,E(e)$ at the delayed index (minus that at the crossing index for an advance) with zero clock cost, so, for $q \neq 0$ and $h \neq 0$, the crossing position is stationary exactly when the seam field vanishes at the reachable indices, and under a static nonzero field it is never stationary and drifts with the sign of qhE. The block momentum $2m((k+1)\tau, s_e)$ is unchanged by every in-block move, which swaps the per-step momenta. Along the respective scaled Ampère evolutions sourced by the original and delayed hopping currents, and with the original field static across the crossing step, the delay difference equals h times the hop's field-energy transfer $(q/2)(E_j + E_{j+1})(e)$ (`TimelikeClassForceLaw`).

The same class meets the field sector's stability window. For nonzero h, uniform boundedness of the electric seam energy for every zero-current datum holds exactly when $h^2(3+\sqrt{5}) < 4$; inside that strict window the magnetic face energy is also uniformly bounded, while equality and larger steps admit an explicit zero-current solution with unbounded electric seam energy. No bound on the gauge potential A follows. A crossing step is timelike exactly when the unit satisfies $\tau^2 > 4$. Under the declared identification of the worldline unit with the field step ($\tau = h$) no step lies in both, so inside the strict energy window every per-step crossing is spacelike and a timelike per-step crossing sits above the threshold with an unbounded electric-energy solution; the two thresholds differ by the factor $2\varphi^2 = 3 + \sqrt{5}$. A block of k rests and one crossing is timelike exactly when $2 < (k+1)\tau$, which inside the window forces $\sqrt{2}\,\varphi < k+1$, at least two rests per crossing, with the two-rest block timelike exactly for $2/3 < h$; the block speed $2/((k+1)\tau)$ is below one exactly on timelike blocks and its supremum over the window is one, approached and never attained. Without the identification the thresholds are independent ($\tau = 3$, $h = 1/2$ gives timelike crossings with uniformly bounded electric seam and magnetic face energies), and the corpus joined witnesses are not identified. The identification is declared and nothing selects it (`SeamStepSpeedLimit`).

Which rule the hosted internal process follows along such a worldline is compared exactly. Index

accrual, one internal step per shared step index at the declared unit, is the per-step member of the internal-action family; proper-length accrual is the declared proper-time principle. Over a window with c crossings the two differ by exactly $E\,c\,(\tau - \sqrt{\tau^2-4})$. For $E \neq 0$ this difference is nonzero on every window with a crossing at every timelike unit and is zero exactly on resting windows; the midpoint refinement of a worldline with a crossing is the generated path of no seam-step worldline. For $E \neq 0$, index accrual fails refinement invariance for every nonzero unit while proper-length accrual passes, so the source chain advancing per shared index is the rule that refinement invariance excludes on every moving worldline with nonzero ledger. The uniform worldline with one crossing per $k+1$ steps has proper length $(k\tau + \sqrt{\tau^2-4})/((k+1)\tau)$ times that of a resting worldline over the same index count, below one for $\tau > 2$, tending to one in k, $\sqrt{5}/3$ at $\tau = 3$, $k = 0$; under the two declared readings of one internal unit the recurrence clock's mean return $61511/7155$ reads as that many shared steps on every worldline, or as the inverse factor times it on the uniform one. Neither rule is selected by the source or the join (`SourceClockRateAlongWorldlines`).

The length rule has a construction. Clock the hosted process by the accumulated proper length in units of a declared internal unit u: the clocked index $c(n)$ is monotone with each increment in $\{0,1\}$ for $\tau \leq u$, equals the shared index on the resting worldline at $u = \tau$, and on the uniform worldline with k rests per crossing obeys $n\,d\tau/u - 1 < c(n) \leq n\,d\tau/u$ on full periods with d the dilation factor, so $c(n)/n \to d\tau/u$, exactly d at $u = \tau$. The clocked return count of the recurrence chain equals the chain's return count at $c(n)$, every per-chain-step quantity converts with the same bound, and the clocked index is invariant under midpoint refinement by construction. No constant step duration of the joined architecture reproduces the accumulated proper length on a moving uniform worldline with at least one rest between crossings; the all-crossing case is matched by the constant duration $\sqrt{\tau^2-4}$, so the clocked chain is a declared enrichment of the join, two internal units give two clocked indices, and nothing selects the unit from the source (`ProperLengthClockedChain`).

The slope with which internal energy enters inertia is then selected by two declared principles, each with an exact theorem. On the momentum map, let the composite four-momentum range over the declared family $P_\lambda(m, E; u) = (m + \lambda E)\,u + (1-\lambda)E\,e_0$, with u on the future unit hyperboloid and e_0 the rest direction; every member has scalar coordinate $m + E$ at the standard frame. Under an oriented Lorentz map L the family transports as $P_\lambda(Lu) = L\,P_\lambda(u) + (1-\lambda)E\,(e_0 - Le_0)$, so P_λ is frame covariant exactly when $\lambda = 1$ or $E = 0$, the converse witnessed by an explicit boost with parameters $(5/4, 3/4)$; the Lorentz square is $(m+E)^2 + 2(1-\lambda)E(m+\lambda E)(u^0 - 1)$, so the shell identity at the rest energy holds at every frame exactly when $(1-\lambda)E(m+\lambda E) = 0$. The selection is kinematic, on the momentum map, and the slope-one member is the composite momentum of the inertia precursor with no factor (`CompositeMomentumCovariance`). On the action, the declared midpoint refinement of a worldline leaves the proper length invariant, doubles the step count, and halves the quadratic clock action; in the internal-action family $E(aL + bM)$ with L the proper length and M the step count, refinement invariance on timelike windows holds exactly when $b = 0$ for $E \neq 0$, which excludes the per-step additive ledger of the second composite shape at nonzero ledger. The proper-time principle, declared, sets the internal action to E per unit proper length; the rest phase of the declared internal clock advances by the mass parameter per unit proper time and its window increment equals that internal action, so the declared internal clock is the proper-length form, and within the family $E\lambda L$ agreement with the principle holds exactly when $\lambda = 1$ for $E \neq 0$. For the length action $(m+E)L$ plus a declared potential pairing, the derivative along every variation exists on timelike windows, and stationarity under endpoint-fixed variations is equivalent to $(m+E)$ times the unit-tangent difference equal to the impulse at every interior node, so the inertial coefficient of the refinement-invariant form is exactly $m+E$; on unit-increment windows the length and quadratic equations of motion coincide (`ProperTimeInternalAction`). The

refinement, the proper-time principle, the momentum family, and the covariance requirement are declared, every shape is a declared enrichment by the realized-history Legendre non-identifiability, and nothing derives the identity; its dynamical content is located in these two principles.

The ledger itself composes exactly. Over a product reference the modular energy of any joint state, correlated or not, is the sum of the marginal ledgers; against a general joint reference the binding defect is the expectation of $\log(\tau_1\tau_2/\tau_{12})$, and against the Gibbs reference of a declared total energy $H_1 + H_2 + V$ it is $\beta\langle V\rangle_p + \log(Z_{12}/Z_1Z_2)$, whose constant vanishes at $V = 0$. For a declared attractive $V \le 0$ and $\beta \ge 0$ the state part is nonpositive while the constant is nonnegative, so the sign of the full defect is not fixed, and a two-point uniform state has the exact positive defect $-1/4 + \log((3+e)/4)$; every normalized faithful joint reference is Gibbs for its own modular Hamiltonian, so the defect is always the expectation of a state-independent effective interaction. Under the declared composite shape the inertial coefficient of a composite is $m_1 + m_2 + E_1 + E_2$ plus the defect, a definitional composition whose physical reading is the calibration import (`ModularEnergyAdditivity`). Matter dynamics beyond these declared shapes and observer readout appear in no field, so the common-world target is unestablished. A physical common action, source-selected carrier identification, matter dynamics, and readout are four named residual categories, not an exhaustive theorem about all required attachments.

5 Histories, Actions and Mechanics

Two standard finite-horizon constructions clarify what the premises A1–A3 must supply before an action principle could be claimed.

Theorem 5.1 (Finite path projection and positive-gap tail)**.** *Let Ω be finite, let $\Gamma = \Omega^{\{0,\dots,N\}}$, and let P, τ be probability laws on Γ, with $\tau > 0$. Let $R > 0$ be a reference weight, let $F_a : \Gamma \to \mathbb{R}$ be finitely many declared observables, and suppose*

$$\log \tau(\gamma) = \log R(\gamma) - \sum_a \lambda_a F_a(\gamma) - \log Z$$

and P and τ have the same F_a-moments. Then

$$D(P\|R) = D(P\|\tau) + D(\tau\|R), \qquad D(\tau\|R) \le D(P\|R).$$

With

$$A_{\text{eff}}(\gamma) = -\log R(\gamma) + \sum_a \lambda_a F_a(\gamma),$$

one has $\tau(\gamma) = \exp[-A_{\text{eff}}(\gamma) - \log Z]$, so every supplied modal history minimizes A_{eff} over Γ.

Separately, for any nonempty finite set Γ, function $S : \Gamma \to \mathbb{R}$, global minimizer γ_0, and

$$q_\beta(\gamma) = \frac{e^{-\beta S(\gamma)}}{\sum_\delta e^{-\beta S(\delta)}},$$

the mass at least $\Delta > 0$ above that minimum obeys

$$\sum_{S(\gamma) \ge S(\gamma_0)+\Delta} q_\beta(\gamma) \le |\Gamma| e^{-\beta\Delta}$$

for $\beta \ge 0$, and this displayed mass tends to zero as $\beta \to \infty$. Finiteness also gives the stronger concentration statement

$$\lim_{\beta\to\infty} \sum_{S(\gamma) > S(\gamma_0)} q_\beta(\gamma) = 0,$$

so the total mass outside the full minimizer set vanishes.

Proof. Expand $D(P\|R) - D(P\|\tau)$. The displayed logarithmic form, normalization, and moment matching make the result independent of P on the constraint surface; evaluating it at $P = \tau$ gives the Pythagorean identity. Nonnegativity of $D(P\|\tau)$ gives minimality. Exponentiating the logarithmic form reverses probability and effective-action order. For the tail bound, the partition sum is at least $e^{-\beta S(\gamma_0)}$, while each displayed numerator is at most that quantity times $e^{-\beta\Delta}$; summing and taking the elementary exponential limit proves the fixed-gap statement. Apply that argument to each strictly nonminimal history and sum over the finite set for the final limit. The statements are checked in Lean [7]; they are instances of standard information-projection mathematics [9] and elementary finite exponential estimates. □

The theorem assumes the exponential form and matching moments. It neither constructs R from the OPH source nor proves that a premise A3 history optimizer exists, has that form, or is unique. Calling R a probability law additionally requires its normalization.

Theorem 5.2 (Real finite-horizon variational helpers)**.** *Let $L : \mathbb{R}^2 \to \mathbb{R}$ be differentiable and define*

$$\mathcal{S}_L(\gamma) = \sum_{n=0}^{N-1} L(\gamma_n, \gamma_{n+1}) \qquad (\gamma : \{0, \dots, N\} \to \mathbb{R}).$$

At an interior junction $k+1$, suppose γ minimizes $\mathcal{S}_L$ under every replacement of γ_{k+1}. Then

$$\partial_2 L(\gamma_k, \gamma_{k+1}) + \partial_1 L(\gamma_{k+1}, \gamma_{k+2}) = 0.$$

If a differentiable one-parameter family T_s satisfies $T_0 x = x$, has generator $\xi(x) = \partial_s T_s x|_{s=0}$, and obeys $L(T_s x, T_s y) = L(x, y)$, then the segment momentum

$$p_n = \partial_2 L(\gamma_n, \gamma_{n+1})\, \xi(\gamma_{n+1})$$

is equal on the two segments adjacent to that junction. More generally, for a path with $M+2$ records, if the same single-site minimization premise holds at all M interior records, then there is a scalar J such that $p_n = J$ on every one of its $M+1$ segments.

Proof. Changing one interior record changes exactly the two adjacent summands. Differentiating their sum at its declared minimum gives the first identity. Differentiating the invariance relation at $s = 0$ and combining it with that identity gives $p_k = p_{k+1}$. Applying this identity at every interior junction and inducting along the finite chain gives the scalar J. These are standard discrete variational and Noether calculations [10]; their precise premises are checked in Lean [7]. □

Proposition 5.3 (Finite/real variation obstruction)**.** *Fix a real path $\gamma : \{0, \dots, N\} \to \mathbb{R}$, a site i, and any finite family G of real paths. There is an $x \in \mathbb{R}$ for which the path obtained by replacing γ_i with x does not belong to G. Hence no finite real-path family contains every single-site real variation of even one supplied path.*

Proof. The map from x to the updated path is injective because evaluation at i recovers x. Its image is infinite and therefore cannot be contained in finite G. This elementary cardinality obstruction is machine checked in Lean [7]. □

The two theorem domains compose only through a certified receipt. The first uses paths in a finite state set; the second uses arbitrary real single-site variations, and Proposition 5.3 rules out closure under all such variations inside any finite real-path family. The composition that survives this obstruction is conditional: an undercut receipt asserts that every real single-site variation of an

embedded path is undercut in action by some embedded finite path, and under that receipt finite minimality transfers to real local minimality and to the real Euler–Lagrange packet. Membership receipts are impossible (that is the obstruction restated); undercut receipts are attainable, and the committed quadratic witness carries one, while a one-path control family provably carries none and the transferred conclusion fails there. All three layers, transfer, witness, and control, are checked in Lean [7].

Which action the history law carries is itself a theorem rather than a choice.

Theorem 5.4 (Derived action of a source chain)**.** *Let Ω be a finite nonempty state set, P a strictly positive row-stochastic kernel, and π a strictly positive normalized initial law. On paths with n transitions define the log-transition action $S_{\log}(\gamma) = -\sum_i \log P(\gamma_i, \gamma_{i+1})$ and the reference $R(\gamma) = \pi(\gamma_0)\,|\Omega|^{-n}$. Then the Markov path law equals the exponential tilt of R by $S_{\log}$ at multiplier one, with partition constant $|\Omega|^{-n}$; and a pair (S', λ') reproduces the path law as its tilt of R if and only if there is a constant c with $\lambda' S' = S_{\log} + c$ pointwise.*

Proof. Exponentiating the negated action recovers the transition product, and row normalization makes the partition sum telescope to $|\Omega|^{-n}$, which cancels against the reference weight. Uniqueness reduces to the statement that two tilts of one strictly positive reference agree exactly when the multiplier-weighted actions differ by an additive constant, proved by taking logarithms at a common path and cancelling the shared normalizer. Both directions of the gauge are realized: per-step additive constants aggregate to a path constant, and the multiplier trades against the action scale, so the bare-action convention chooses $\lambda = 1$ as a representative. It is the unique multiplier for that unrescaled action when the path action is nonconstant; for a uniform kernel the action is constant and normalization erases the multiplier. The statements are checked in Lean [7]. □

The committed source chain instantiates the theorem exactly: the tilt of its reference by its log-transition action reproduces its path law with partition constant $1/4$, through kernel-decided integer identities, and the committed repair-count action reproduces that law at no multiplier, because it takes equal values on two paths that the reference weighs equally and the chain separates. The action of the realized dynamics is therefore forced up to an additive constant and a multiplier rescaling. This multiplier is an action-normalization gauge for the supplied chain; it is not thereby identified with a gauge-sector coupling coefficient. The separate finite Gibbs-kernel construction for one three-dimensional P-family factor shows that a common rescaling of its supplied quadratic cost can occupy the same formal multiplier slot, but it neither derives that kernel from source dynamics nor determines relative couplings between simple factors.

The retained length-three history packet also has an exact finite operator attachment. Let $p_g = n_g/1754$ be its empirical law on the eight binary paths and let $S_g \in \{0, 1, 2\}$ count the two adjacent state changes. The diagonal matrices

$$\rho_{\text{hist}} = \operatorname{diag}(p_g), \qquad H_{\text{hist}} = \operatorname{diag}(S_g)$$

form a strictly positive normalized density and a positive non-scalar self-adjoint Hamiltonian, with

$$[\rho_{\text{hist}}, H_{\text{hist}}] = 0, \qquad \operatorname{Tr}(\rho_{\text{hist}} H_{\text{hist}}) = \frac{197}{1754}.$$

On a one-regulator consensus tower, the selected-state construction places this pair on its completed-colimit GNS Hilbert space. The finite-stage representation is injective, the represented Hamiltonian and one off-diagonal commutator response are nonzero, and the mapped exponential flow is unitary and leaves the density stationary. The cyclic unit has nonzero energy, so it is not a ground state or vacuum for this Hamiltonian.

The eight histories also reindex exactly as three binary tensor slots. Their one-slot algebras and the two adjacent interval algebras are isotone, the first and last slot algebras commute, and the Hamiltonian splits without a remainder into positive local bonds,

$$H_{\mathrm{hist}} = H_{01} + H_{12}, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{01}) = \frac{94}{1754}, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{12}) = \frac{103}{1754}.$$

The first commutator of an endpoint observable lies in its adjacent two-slot algebra. The source functional restricts compatibly along all these algebra inclusions, and the locality, bond, and propagation identities survive the injective selected-GNS representation. This supplies a finite history-coordinate local dynamical system on a selected-GNS carrier built from the source-counted diagonal functional. The retained source packet fixes the diagonal probabilities and repair counts. The extension to the full matrix algebra, the tensor-local reading, off-diagonal observables, and the real flow parameter are mathematical adapters. The history slots are not calibrated physical times or Lorentzian regions, and no spectrum condition, continuum/RG limit, physical field, particle, scattering, or detector interpretation is obtained from them [7].

The rational record layer of the same carrier carries a finite-probability conditioning family. Conditioning the window law on any slot region defines a rational-linear, unital, pointwise-positive, localized, idempotent, tower-compatible map that preserves the empirical mean; the walls are fixed points of their own regional maps, and the exact conditional weights of the late slot are 383/415 and 2/89 in two (s_0, s_1) cells of the counts 1149 : 96 and 2 : 87. These counts give a narrowly scoped matrix no-go: no complex-linear map from the full matrix algebra into the fixed first two-slot algebra $\mathcal{A}_{01}$ can both obey the $\mathcal{A}_{01}$-bimodule law and preserve the committed state. They do not rule out ordinary AQFT inclusions with state restriction, a different or enlarged target algebra, ancillary boundary memory, generalized expectations, or non-bimodule positive or completely positive channels [7]. The enlarged-target path is constructed: a unital injective star-embedding reads the slot-two record bit of each history into one qubit ancilla, its range is exactly the ampliated interval algebra joined with the ancilla factor, and it satisfies clause for clause the conjunction the fixed-target no-go kills, with the range clause the single clause whose target changes and the exhibited state a proved non-product extension restricting to the committed source state. Two delimiting theorems establish two necessary features of this escape: a product-form state extension keeps the obstruction for every ancilla matrix, and a map into the ampliated copy alone is obstructed even against the correlated extension. Thus, within this clause family, the ancilla factor and some non-product state extension are necessary. The exhibited record-correlated extension is one witness and is not proved unique. This is a finite-matrix existence statement; regions, time slices, dynamics, and any continuum attachment are not touched by it [7].

The same packet also fixes a bounded overlapping-window action statistic. Among 1754 retained length-three windows, the two domain-wall indicators have count-weighted incidences 94 in the first edge position and 103 in the second, hence mean two-edge action 197/1754. The value 197 counts incidences inside overlapping windows, not distinct underlying repair events. The formal object stores only the eight-path histogram and chosen observables: it contains no window order, successor relation, cadence, or period. Dividing by an externally declared positive real, interpreted as a window duration, is therefore only a conditional unit conversion; that interpretation is not related to the histogram by any theorem. This supplies neither a source-derived clock nor a reduction of the physical calibration input [7].

The time row is located more exactly by two finite results. Accepted repair hosts no periodic clock: every accepted step strictly lowers the mismatch count, so the accepted-repair relation admits no cycle, and the canonical repair iteration admits no periodic orbit other than a fixed point. The committed stationary source chain, by contrast, carries a recurrent process: the first-return law of

each state is written in closed form from the kernel, sums to one, and satisfies Kac's identity with the exact mean return times 61511/7155 and 61511/54356 steps, whose ratio 54356/7155 is the same number under every declared tick. The return law assigns positive probability to both one- and two-step returns. Its standard Markov-chain period is therefore one, while no deterministic return interval exists because no return time has probability one. A return count along a trajectory counts distinct visit events, the property the overlapping incidence statistic lacks. One chain step stays a declared tick and no physical duration is identified; the abstract rate non-identifiability result concerns transition systems whose bridge to the repair layer fails, and it forbids nothing here (`SourceRecurrenceClock`).

The field sector, by contrast, hosts periodic processes. Every eigenvector v of the local operator $C^{\mathsf{T}}C$ with eigenvalue λ carries, at a declared nonzero step h with $0 \le h^2\lambda \le 4$, the histories $A_n = \cos(n\theta)\, v$ and $A_n = \sin(n\theta)\, v$ with $\theta = \arccos(1 - h^2\lambda/2)$; they solve the zero-current scaled evolution in the temporal gauge, conserve the staggered form, and are bounded. Kernel-checked integer eigenvectors with eigenvalues 2 and 3 join the committed ones with eigenvalues 5 and $3+\sqrt{5}$. At $h^2 = 1/2$ the eigenvalue-two mode returns after six steps, at $h^2 = 2/3$ the eigenvalue-three mode after four, at $h^2 = 1/3$ after six, each step inside the sharp window $h^2 < 2/\varphi^2$, since $3 \pm \sqrt{5} = 2\varphi^{\pm 2}$; the projector traces $1, 5, 4, 4, 6$ read with rank equal to trace give the multiplicities $1, 5, 4, 4$ and $3+3$. Frequency ratios carry no tick: $h\sqrt{\lambda} \le \theta \le h\sqrt{\lambda}/\sqrt{1 - h^2\lambda/4}$, $\theta/h \to \sqrt{\lambda}$ as the step shrinks, and the small-step ratio of the golden mode to the slowest mode is φ^2 exactly. A period p and the mean return time 61511/7155 of the recurrence clock are counted in one shared step index, with ratio $7155p/61511$ under every declared tick. The step, the gauge, and the kinetic term are declared; the repair-layer no-periodic-orbit theorems and these field-sector periods have disjoint scope (`CarrierModeOscillators`).

The discrete step embeds in a continuous flow. On the amplitude-velocity plane of every eigenvector v with eigenvalue λ and $0 < h^2\lambda < 4$, the zero-current temporal-gauge evolution is the symplectic map $T = \left(\begin{smallmatrix} 1 & h \\ -h\lambda & 1 - h^2\lambda \end{smallmatrix}\right)$ of determinant one and trace $2 - h^2\lambda$, conjugate by an explicit matrix Q to the rotation by the mode angle; the family $\Phi_t = Q\,\mathrm{rot}(\theta t/h)\,Q^{-1}$ is a continuous one-parameter group with $\Phi_h = T$ and $\Phi_{nh} = T^n$ that preserves the committed staggered energy, equal to $(\|v\|^2 \sin^2\theta/2)$ times the pulled-back Euclidean form. On the kernel of the local operator, exactly the gradient space, the evolution is the h-step of the shear flow. For any finite family of pairwise orthogonal admissible eigenvectors the componentwise coefficient-state flow is a linear continuous one-parameter group whose h-step is the committed step, instantiated on the eigenvalue-two, three, five, and golden modes and a gradient. The assembled theorem permits zero listed vectors, so a faithful flow on the actual field span additionally requires nonzero linear independence or quotienting redundant coefficient directions. This generic flow theorem alone supplies neither an inner product nor the full-curl selection; the energy construction and explicit nineteen-vector basis below supply those separate inputs. Reading t as physical time remains declared (`CarrierEvolutionFlow`).

Inside the strict Courant window $0 < h^2\lambda < 4$ the staggered energy of a carrier mode is a positive definite quadratic form on the state (a, b), and fails to be one exactly at and beyond the boundary: at $h^2\lambda = 4$ the state $(-h/2, 1)$ has zero energy, beyond it negative energy, and on the gradient sector the amplitude direction is null. Its polarization is an inner product on each curl mode, the committed step and the interpolating flow are isometries, and for a nonzero step every step-invariant symmetric bilinear form is a real multiple of it, the committed energy fixing the multiple. The conjugated quarter turn squares to minus one and is orthogonal for this inner product, so each mode carries a complex line on which the flow is the phase $e^{i\theta t/h}$, with orbit derivative $i(\theta/h)$ times the state; the generator is exhibited, not obtained from Stone's theorem. On a finite orthogonal family of curl modes the direct sum is an inner product whose diagonal

is the committed staggered energy at every step, the assembled flow is unitary with diagonal generator θ_i/h. Precisely the static gradient-amplitude states with zero electric component form the radical of the semidefinite extension; gradient velocity/electric directions are non-null. On an equal-eigenvalue block invariance alone does not fix the form and the energy does. The step h is a declared Courant number and t a declared flow parameter (`FieldSectorEnergyInnerProduct`).

The curl sector of the committed seam operator has an explicit orthogonal eigenbasis: nineteen seam vectors with entries in $\mathbb{Z}[\varphi]$, five at eigenvalue 2, four at 3, four at 5, three at $3+\sqrt{5}$ and their Galois conjugates at $3-\sqrt{5}$, kernel-checked as eigenvectors with diagonal Gram matrix. The four committed modes are rows of the list. The nineteen span the range of the codifferential, which is the kernel of the boundary; eleven coboundaries of listed port loads span the gradient sector, and the two sectors are complementary. On this basis the coefficient flow extends to the whole seam space: for a declared step with $h^2(3+\sqrt{5}) < 4$, the two-sided admissibility window of the family, every zero-current temporal-gauge solution is the assembled history of its initial state. On all nineteen nonzero curl modes, the weighted energy form is positive definite, its real and Hermitian forms are flow-invariant, its complex coordinates have the explicit phase generator, its diagonal is the field energy, and the coefficient state is recoverable from potential and electric readouts. After conjugating the coefficient orientation, the packet has an exact finite Stone representation for the selected principal-angle interpolation on a separately constructed $M_{19}(\mathbb{C})$ block: the self-adjoint diagonal Hamiltonian has $H_{ii} = \theta_i/h$, its propagator is e^{-iHt}, and conjugation of the coefficient outer product is exactly the outer product of the evolved coefficient state. The vector intertwiner retains absolute phase; the rank-one outer product loses global phase and retains relative phases only. The energy-weighted outer product has trace equal to the assembled field-energy Hermitian norm. A selected mode is driven by the joined scalar clock at discrete steps under the sufficient calibrations $\delta = h$ and $m = \theta_i/h$. The 19-index block is not a relabelling of the committed $M_2(\mathbb{C})$ private carrier, the committed witness fails the duration calibration, and the eigenvalue-two and eigenvalue-three principal-branch frequencies are unequal throughout the strict positive window. Thus no one identical scalar rate equals every selected principal-branch mode frequency. That statement does not exclude a shared time parameter, phase aliases, harmonics, finite-horizon decoding, mode-resolved rates, or operator-valued readout. There are five distinct frequencies among the nineteen coordinates; the logarithm branch is selected rather than uniquely derived from the discrete step, and the two displayed calibrations are sufficient rather than necessary modulo 2π aliases. The coefficient input is arbitrary and is not proved to be the joined record's actual Maxwell history. The rank-one projective image is neither onto $M_{19}(\mathbb{C})$ nor an algebra homomorphism, and it is not an equivalence with the committed private algebra. A physical bridge therefore requires a specified compression/channel, a selected lower-dimensional sector, or an enlarged private carrier. This is not a source-selected public representation or Hamiltonian, physical time calibration, photon Hilbert-space identification, or laboratory Schrödinger law. The step and flow parameter remain declared (`CurlSectorEigenbasis`, `CurlStoneClockBridge`).

These modes are sorted by the carrier's own symmetry. The sixty listed icosahedral port permutations induce signed permutations of the seams and orientation-preserving permutations of the faces, the signed face–seam incidence is invariant under the simultaneous action on all $60 \times 20 \times 30$ triples, and so the curvature, the codifferential, the local operator, the five spectral projectors, the mode histories, the scaled Ampère evolution for every step, potentials, and current, and the staggered energy commute with the action, all by kernel checks on the committed tables (the intertwining of the operators, fields, evolution, and energy under the same sixty rows was proved with the carrier dynamics compatibility; new here are the explicit signed actions with their composition laws, the projector and eigenspace invariance, the mode transport, and the characters). The constant face vector spans the fixed eigenvalue-zero line, the element orders are $1, 2, 3, 5$ with

class sizes $1, 15, 20, 24$, and the traces of the action on the projector images by element order are $1;\ 5, 1, -1, 0;\ 4, 0, 1, -1;\ 4, 0, 1, -1;\ 6, -2, 0, 1$, with character norms $60, 60, 60, 120$. Read with the orthogonality relations, an inference outside the checked statements, the face space decomposes as $1 \oplus 5 \oplus 4 \oplus 4 \oplus (3 \oplus 3')$, the two four-dimensional eigenspaces carrying one irreducible representation and the golden sector the two three-dimensional ones; the checked characters do not separate the two order-five classes, so the split of the golden sector by eigenvalue is an observation. The field modes are thus sorted by the same group that sorts the matter sectors, with the identification of a mode with a physical oscillation open (`CarrierModeEquivariance`).

The finite six-axis interface also has an abstract group endpoint. The antipodal port bridge is first a pointwise equivalence of indexed action rows. Separately, the canonical center quotient $SL(2, \mathbb{Z}/5) \to PSL(2, \mathbb{Z}/5)$, whose kernel is the center $\{+I, -I\}$ of $SL(2, \mathbb{Z}/5)$, yields a faithful action of $PSL(2, \mathbb{Z}/5)$ on the projective line under the explicit coordinates $[z : 1] \leftrightarrow z$ and $[1 : 0] \leftrightarrow 5$; its image is exactly the committed sixty-element subgroup `A5SixAxes.L60`, and the result is a typed group isomorphism onto that subgroup (`A5PortSixAxesBridge`, `PSL2F5SixAxesBridge`). The sixty committed twelve-port rotations are then packaged as a typed subgroup of the permutations of the twelve ports, with the multiplication and inverse laws checked on the actual port permutations, and the pointwise port bridge is promoted to a group isomorphism between that port subgroup, the six-axis subgroup, and $PSL(2, \mathbb{Z}/5)$. The two golden pieces are then carried as typed representations: the left face action is packaged as a linear representation of the port subgroup on the twenty-dimensional face space, restricted to either rank-three golden projector image, transported along the isomorphism to $PSL(2, \mathbb{Z}/5)$, and pulled back through the center quotient to $SL(2, \mathbb{Z}/5)$, where every central element, in particular $-I$, acts trivially; the row law behind the representation is an exact finite-table check against the certified port-group multiplication (`GoldenSectorPSL2F5Representations`). This is not an abstract $PSL(2, 5) \simeq A_5$ classification, an identification with the binary icosahedral group or McKay E_8, a two-dimensional spinor representation of $SL(2, \mathbb{Z}/5)$, a selection of φ or a mass law, or a physical-rotation statement.

The golden split is exact. On the six-dimensional golden sector, where $N^2 - 6N + 4 = 0$, the two spectral projectors with eigenvalues $3 \pm \sqrt{5} = 2\varphi^{\pm 2}$ have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, are symmetric orthogonal idempotents summing to the golden projector, each of trace 3, and the minus projector is the entrywise Galois conjugate ($\varphi \mapsto 1 - \varphi$) of the plus one. On the sixty listed automorphisms the plus character takes the values $3, -1, 0$ on the orders $1, 2, 3$ and φ or $1 - \varphi$ on the twenty-four elements of order five, twelve each; the minus character is its Galois conjugate on every element, the two agree exactly off order five, squaring an order-five element exchanges them, both are class functions commuting with every automorphism, and each has norm 60. Read with the orthogonality relations, these data have the usual two-character interpretation, but this character module does not itself prove irreducibility. The separate span certificate below proves that the two real projector images are three-dimensional and irreducible for the listed incidence action. Scalar extension to complex irreducibility, identification with the abstract icosahedral or A_5 character table, and a physical-oscillation reading remain outside the checked statements (`GoldenSectorCharacters`).

Irreducibility of the two golden pieces is a theorem. Each image $W_\pm$ of the golden projectors is invariant under every listed automorphism and has dimension three (an explicit factorization $P = AB$ with $BA = 1$); the real span of the sixty matrices $gP_\pm$ has dimension nine and maps onto all 3×3 real matrices in the basis A (nine listed elements with an explicit two-sided inverse certificate over $\mathbb{Z}[\varphi]$, the minus certificates the Galois conjugates of the plus ones), so every invariant subspace of $W_\pm$ is zero or all of it: both pieces are irreducible real representations of the listed incidence-automorphism action (`GoldenSectorIrreducibility`). The pieces stay irreducible after extension of scalars to $\mathbb{C}$: the nine listed automorphisms whose transported tables span the real endomorphism algebra span the complex one with the same inverse-certificate coefficients, so every

invariant complex subspace is zero or the whole piece. The two complexified pieces are inequivalent, since on the listed row of order five their traces are $1-\varphi$ and φ, every complex intertwiner between them is zero, and the real endomorphism algebra of each piece is the real scalars, so both are of real type (`GoldenSectorComplexIrreducibility`). Identification with the abstract icosahedral or A_5 table, with physical rotations, or with physical oscillations is unproved.

The two attained faces of mechanics are joined by a finite Legendre bridge, also checked in Lean [7]. For discrete two-point Lagrangians the Legendre transform, its involutivity on the strictly convex class, the Fenchel–Young inequality with its equality case, and exact degenerate controls are proved; the discrete Euler–Lagrange condition at a junction holds exactly when one step of the discrete Hamilton flow carries the incoming junction state to the outgoing one, for the quadratic class and for strictly convex Lagrangians with a solver section; single-site minimizers of the log-transition local action coincide with most probable paths of the realized chain, through an exact corner identity; and the constant Noether current of the chain face equals the quadratic Legendre momentum, conserved together with the energy along the free Hamilton orbit that reproduces the committed witness path. One remainder is named: no theorem produces the realized chain from a Hamiltonian flow. What is proved in that direction is narrower. The committed repair-count action reproduces the chain path law at no multiplier and lies outside the additive-constant and rescaling gauge orbit of the derived log-transition action (`sourceRepairAction_no_multiplier`, `sourceRepairAction_not_gauge`), so the naive quadratic-Gibbs identification is refuted on the chain literals, and the derived action above is the only route this package supplies. A direct Hamiltonian-flow derivation of the chain, for example a block Hamiltonian of the Stone converse coarse-grained to the two-state chain, is excluded by no theorem here and is unconstructed.

Theorem 5.5 (Realized histories do not select a real Legendre enrichment)**.** *Let $L_0(x,y)$ be the bilinear real extension of the committed two-state log-transition table. It is affine in y, so its momentum at fixed x has singleton image and admits no global velocity solver. For every $a \in \mathbb{R}$, define*

$$L_a(x,y) = L_0(x,y) + \frac{a}{2}\, y(y-1).$$

Every L_a agrees with the exact source log-transition action on every realized binary history at every path length. For $a > 0$, however, L_a is strictly convex in y, has an explicit global velocity solver and Legendre transform, and different positive a's give different formulas. In particular the checked $a = 1$ and $a = 2$ members have distinct Lagrangians and Hamiltonians, agree on all source histories but differ by $1/8$ when one middle record is varied to $y = 1/2$.

Proof. The correction $y(y-1)$ vanishes at both source symbols. Differentiation gives $\partial_y L_a = \partial_y L_0 + ay - a/2$, so $a = 0$ is degenerate and $a \neq 0$ has the displayed affine inverse. Completing the square gives the Legendre transform; the strict-convexity gap is $(a/2)b(1-b)(y-z)^2$. The path identity, midpoint control, solver, convexity, and the $a = 1, 2$ distinction are checked in Lean. □

The reference measure is characterized within a declared independent-target scrambling normal form: among row-stochastic kernels, invariance under every independent relabeling of transition targets at fixed source, row-constant transition weight, and constant log-transition step action are each exactly equivalent to the uniform kernel, so any independently target-relabeling-invariant Markov reference has the step-uniform path law, and a biased two-state control shows the invariance premise is load-bearing. Ordinary simultaneous source-target relabeling is strictly weaker and admits a positive nonuniform stay-biased control. The stronger condition fixes the reference gauge by an invariance convention, exactly as a coordinate convention is fixed; the invariance principle itself is stated, not source-produced. Feeding the separately declared counting reference and trivial

visible datum into the premise A3 conditional-resampling construction reproduces the uniform transition kernel exactly. This is a conditional realization, not source selection of those inputs, the initial distribution, or the complete path reference; constant rescalings of the counting mass are another exact ambiguity. The committed reference/action/multiplier receipts also belong to two distinct Gibbs constructions and do not form one source-selected capstone. Within the repair-count construction, the matching polynomial is strictly increasing on the positive exponential-parameter ray, so the supplied empirical target has one unique matching parameter. The target observable and level are declared postprocessing inputs rather than prospectively source-selected constraints.

A three-record concave-action control has a stationary history that is not a minimum and is assigned less positive Gibbs weight than a unit variation. It proves that a positive Gibbs/MAP rule need not make every stationary point a mode or minimizer. It is not a no-go for constrained saddles, complex stationary phase, path-integral interference, or a source-selected real Lagrangian, all of which remain available. The package identifies neither A_{eff} with the local action nor the finite scalar chain constant with a physical current, and supplies no amplitude, clock, or continuum path measure. The theorem above also shows that the realized two-state history law, on the committed chain literals, does not select the off-alphabet curvature needed for a regular Legendre system; finer alphabets of the same run, refinement limits, other source data, and declared principles are untested by that theorem and stay live routes to the curvature. It derives the finite action and conditional bridges at the representation level; the real enrichment and physical attachments are not supplied under the attachment doctrine.

Two composition results sharpen this boundary. First, the two halves of the mechanics package form one theorem (`source_to_hamiltonian_composed`): under the registered path reference, real enrichment, and supplied-dynamics rows, a fixed-endpoint real-extremal embedded history realizes the derived corner action of the same kernel whose exponential tilt over the registered reference is the path law, is an interior most-probable update of that law at every interior site, and satisfies the discrete Hamilton equations. The separate `source_to_hamiltonian_composed_noether` theorem makes the segment momenta constant under supplied symmetry data; the mode/minimizer scoping and the enrichment no-go ride along as conjuncts of the same statement. Second, while the realized law cannot select the enrichment, a declared mode-extremality principle fixes one member inside the registered one-parameter curvature ansatz on the committed chain: interior stationarity of the embedded constant-record history forces the curvature $a_\star = 2\log(W_{11}^2/(W_{10}W_{01})) > 0$, positivity being exactly strict interior mode dominance, and at $a_\star$ that history is a genuine fixed-endpoint single-site minimizer at every interior junction and length, so the composed conditional is inhabited on the committed kernel rather than vacuous. This is not uniqueness among regular corner-preserving enrichments, and the residual freedom is exactly characterized inside one declared grammar, in the module `EnrichmentCharacterization`. Within the quadratic two-slot grammar, a sentence vanishes at the four binary corners exactly when it is $\varepsilon\, x(x-1)+\zeta\, y(y-1)$, so the corner-invisible enrichments are exactly the two-parameter family $L_{a,c} = L_0 + a\, y(y-1)/2 + c\, x(x-1)/2$, with injective parameters and no third corner-invisible quadratic direction (`quadPoly_cornerInvisible_iff`). The committed interior-junction stationarity holds on this family exactly on the affine line $a+c = a_\star$ (`chainTwoSlot_stationary_line`); members of that line share the binary corners, the constant-history stationarity, and the single-site variation minimum, while the regular member $a = c = a_\star/2$ has a different velocity curvature. One named velocity-only clause, that the added sentence is a function of the velocity slot alone, restores uniqueness at $(a, c) = (a_\star, 0)$, the committed one-parameter rule with all six grammar coefficients pinned (`quadPoly_selection_unique`). Each clause is load-bearing: the counterfamily point $a = c = a_\star/2$ satisfies every clause except velocity-only and carries a different momentum map (`velocityOnly_clause_necessary`); an explicit velocity-only cubic satisfies corner invisibility and the junction stationarity equation, so quadraticity is load-

bearing under stationarity alone (`quadraticity_clause_necessary`); and the committed real fixed-endpoint single-site minimality excludes those cubics and returns the committed rule exactly (`chainCubic_realMin_forces_quadratic`). This cubic exclusion is not global selection. For every $0 < \lambda < a_\star$, the velocity-only quartic continuation

$$L_\lambda(x, y) = L_{a_\star}(x, y) + \lambda y^2(y-1)^2$$

is corner-invisible, has the same junction stationarity, and keeps the constant-one history globally minimal under every fixed-endpoint single-site variation, with exact gap $(x-1)^2(a_\star/2 + \lambda x^2)$. Its velocity Hessian is $(a_\star - \lambda) + 12\lambda(y - 1/2)^2 > 0$, so its momentum map is strictly increasing, surjective, and globally invertible, yet it is neither the committed rule nor any member of the quadratic grammar (`quartic_continuation_blocks_global_selection`). Thus the tested clauses are not a proved-minimal or globally selecting axiom system.

The quartic boundary is sharpened by one further named clause, tested in `PolynomialEnrichmentSelection` against the full two-slot polynomial grammar at every degree bound. A two-point sentence is a discrete null Lagrangian when no single-site interior replacement changes the sum of the two adjacent terms; by the committed difference identity such sentences contribute only telescoping and boundary terms to every fixed-endpoint local action. The literal boost clause, that every shift residue $q(x+u, y+v) - q(x, y)$ is null, fails at the committed increment itself (`committed_increment_not_boostNull`): a lattice-anchored quadratic has a linear, non-telescoping boost residue. The named candidate clause is second order: every boost difference of a boost difference is null (`BoostNullCovariant`). Every quadratic-grammar sentence satisfies it. Among velocity-only polynomial enrichments of every degree bound the clause holds exactly for the quadratic-affine members (`velocityPoly_boostNull_iff`), and the quartic continuation fails it at an explicit finite witness (`quarticIncrement_not_boostNullCovariant`). In the full polynomial grammar, corner invisibility, the velocity-only clause, the committed stationarity calibration, and the boost clause select exactly the committed point (`polyGrammar_selection_eq_committed`), and the committed point satisfies all four clauses, so the axiom system is inhabited. Every clause is load-bearing: dropping the boost clause readmits the quartic, and in the full two-slot grammar the boost clause does not force quadraticity, since the antisymmetric cubic $(x-y)^3 - (x-y)$ is corner-invisible, satisfies the boost clause, is neutral for the committed stationarity calibration, and equals no sentence of the committed quadratic grammar (`boost_clause_does_not_force_quadraticity`); the velocity-only clause is what removes this direction. The boost clause is a declared candidate axiom rather than a derived symmetry: no boost action on realized histories and no invariance of the source law is constructed, and smooth non-polynomial enrichments are outside the theorem.

The grammar, the mode-extremality principle, and the velocity-only clause are declared rather than source-derived; nothing here states that the source selects the enrichment, and no real enrichment is supplied. The realized history law alone selects no real enrichment (Theorem 5.5).

6 Thermodynamics from Conditional Repair

On one finite regulator stage, the four laws form a conditional theorem package once one repaired distinction and two compatible instantiations of premise A3 are supplied. Five named source and physical receipts are unsupplied. The Lean development carries three composed named-premise surfaces for these packages. `FourLawAdequacySurface` proves one theorem, `fourLaws_composed`: one typed antecedent bundle whose fields are exactly the registered repair law, the clock-and-energy calibration, and the refinement-uniform gap family, and one conclusion record whose clauses all

consume that bundle: the kernel receipts, both zeroth-law forms, the modular and diagonal-heat-channel first laws, the second-law and fluctuation statements, the finite and refinement-uniform third laws threaded to the same calibrated energy through an explicit member identification, and the Landauer bound. The same surface carries an exact dissipation identity: the mean fluctuating entropy production of one repair step equals the relative entropy from the input state to its repaired image, is strictly positive whenever the step changes a strictly positive state, and vanishes exactly on the fibre-conditional-reference fixed points, so one state-changing projection has a strictly positive dissipation. Because this repair kernel is idempotent, repeating the identical step stalls on the fixed manifold after that first projection; a sustained macroscopic arrow requires source dynamics. `HorizonThermalitySurface` carries the cap Clausius inequality and the capacity identities as the horizon-facing composition. `MechanicsAdequacySurface` re-exports the composed source-to-Hamiltonian theorem of Section 5. Three maps stay distinct. The strict-descent normalizer settles public facts; it is terminating and idempotent and carries no entropy inequality, and the two-point normalizer $a \mapsto b$, $b \mapsto b$ sends the uniform law to a deterministic law, dropping Shannon entropy from $\log 2$ to zero. The equilibrium repair operator resamples unresolved degrees inside the fibre of the complete repaired visible datum. The modular flow of a faithful reference supplies the reversible observer ordering and preserves relative entropy. Only the middle map carries the arrow.

The refinement-uniform gap family is the module `CofinalSpectralTailFamily`. It carries per-stage finite spaces and energies, refine maps obeying identity, composition, energy and ground-compatibility laws, one uniform spectral-tail envelope, a nonempty ambient regulator preorder, an order-reflecting cofinal stage embedding, no maximal ambient regulator, and the data-level condition

$$\forall r\ \exists s > r, \qquad |X_r| < |X_s|.$$

The represented stage type is therefore infinite, and a constant finite carrier cannot inhabit the interface; a nonconstant $\mathbb{N}$-indexed ladder with strict carrier growth supplies a nonvacuous inhabitant. The envelope gives one inverse-temperature threshold per tolerance for the off-minimum Gibbs mass uniformly over all stages and, for every ambient regulator, over the complete cofinal tail above it (`uniform_concentration`). The ladder family also proves that a spectral-tail envelope does not imply a uniform total-cardinality bound on the same carriers and energies (`ladderFamily`, `ladder_uniformGap_isEmpty`). The theorem `fourLaws_composed_cofinal` constructs the refinement-independent finite-law core directly from the declared repair and calibration, and every regulator-family conclusion refers to the same supplied cofinal family. A separately supplied calibrated-stage energy identification yields exact off-minimum-mass equality and the same envelope at that member; the calibration is not derived. Strict carrier-cardinality growth is one sufficient genuine-refinement route, not a necessary one. No OPH source produces this family or identifies its regulator, energy, inverse temperature, repair or continuum meaning.

Premise A3 applies separately to states and to transition distributions, and both instantiations are solved exactly. On the state side, with faithful global reference σ_0 and observer-visible conserved quantities Q_a, assume the constrained moment surface contains a faithful feasible state. On this interior branch, the information-projection Pythagorean identity $D(\rho\|\sigma_0) = D(\rho\|\tau_\lambda) + D(\tau_\lambda\|\sigma_0)$ holds on the constrained moment surface, so the optimizer, unique by strict convexity of relative entropy, has the finite-multiplier exponential form $\tau_\lambda = Z^{-1}\exp(\log\sigma_0 - \sum_a \lambda_a Q_a)$; on the tracial branch this is the Gibbs family. Without the faithful-feasibility premise, a boundary constraint can force a rank-deficient optimizer with no finite full-algebra multiplier; it needs a support-restricted or limiting treatment. The exact zeroth-law receipt is temperature identifiability: on one nondegenerate finite spectrum, two Gibbs distributions that agree have equal β. It is not a contact-dynamics or equilibration theorem. On the transition side, the feasible set at x is the fibre

$F_C(x) = \{y : b_C(y) = b_C(x)\}$ of the complete repaired visible datum b_C, and the minimizer of $D(q\|\pi)$ over the fibre, unique by the same strict convexity, is weighted conditional resampling,

$$P_C(x,y) = \mathbf{1}_{\{b_C(y)=b_C(x)\}} \frac{\pi(y)}{\pi(F_C(x))},$$

which is the weighted observation-fiber resampling projector, and any physical identification of a concrete repair law with it passes through the support, equal-fiber-row, and weighted detailed-balance receipt of the observation-normal-form construction [11]. The kernel is stochastic, idempotent, reversible, and stationary for π, and fixes every fibre-measurable observable, which is dynamical conservation of every protected charge and the content side of the first law. For a finite joint update, the exact bookkeeping identity is $\Delta U = \delta Q + \delta W + \operatorname{Re}\operatorname{tr}(\delta\rho\,\delta H)$. The two-term form is first-order, or exact for an ordered stroke with one of H or ρ held fixed; the composed repair heat stroke holds H fixed, so its work and cross terms vanish. Data processing then gives the transition-side second law $D(pP_C\|\pi) \le D(p\|\pi)$, and the exact split $D(p\|\tau) = \langle K\rangle_p - S(p)$ with $K = -\log\tau$ turns it into the modular Clausius inequality $\Delta S \ge \Delta\langle K\rangle$ for every τ-preserving repair channel, which specializes to $\Delta S \ge \beta Q$ on the identified energy branch. The Landauer bound is a one-line corollary: a repair channel that lowers the record entropy by c expels at least c/β of energy, so erasing one bit costs at least $k_B T \log 2$. At every finite regulator the excited Gibbs mass obeys $q_\beta \le \frac{d-g_0}{g_0} e^{-\beta\Delta}$, with the quantitative threshold that any target mass ε is reached once $\beta\Delta$ exceeds $\log\frac{d-g_0}{g_0\varepsilon}$, and the entropy limit $k_B \log g_0$ is the standard finite consequence of the displayed mass bound and finite continuity of entropy; the composed Lean record contains the mass bound, not a separate entropy-limit field. The repair kernel preserves full support, since every diagonal entry is positive, while the zero-temperature state is rank deficient for $g_0 < d$; finitely many repair or pinching steps therefore cannot reach it, which is the fixed-regulator third law. Public quantum-record pinching and averaging have explicit normalized Kraus families and preserve trace. A separate module proves these channels completely positive and trace preserving through the Kraus criterion; this result's own packet reports Kraus and trace identities. The record tower $S(\rho) \le S(\mathcal{E}_\pi\rho) \le S(\mathcal{A}_\pi\rho)$ separates conditioned learning from the nonselective commit, so observers convert uncertainty into correlations and durable records without violating the second law.

The second law refines to an exact identity. For one repair step the fluctuating entropy production $\sigma(x,y) = \log\frac{p(x)}{\pi(x)} - \log\frac{(pP_C)(y)}{\pi(y)}$ obeys three exact relations. The integral identity $\langle e^{-\sigma}\rangle = 1$ holds for every kernel that preserves the reference; under detailed balance the pointwise relation $p(x)\,P_C(x,y) = e^{\sigma(x,y)}\,(pP_C)(y)\,P_C(y,x)$ holds at every pair of states; and summing it over a level set gives the forward weight at entropy production s as e^s times the reversed weight. The mean of σ is exactly the certified descent $D(p\|\pi) - D(pP_C\|\pi)$, so the second law is the Jensen shadow of an identity rather than a primitive inequality. Detailed balance also gives finite Onsager reciprocity. Because $e^{-\sigma}$ is a ratio of rationals, the fluctuation certificate closes exactly over the rationals, with no numerical logarithm.

The reversible symmetry and correlation representation below are finite relatives of standard Onsager and Green–Kubo theory [12, 13]. The OPH-specific statement is the resulting model-selection obstruction for the full-fibre publicization projector.

Theorem 6.1 (Finite transport kernel and repair-projector obstruction)**.** *Let K be a finite stochastic kernel reversible under a strictly positive weight π, let $L = I - K$, and let R be a linear solver of $LRj = j$ on the π-centered currents. For any finite current family $\{j_a\}$,*

$$\Gamma_{ab} = \langle j_a, Rj_b\rangle_\pi$$

is symmetric and positive semidefinite. At every finite cutoff N,

$$\langle j, Rk\rangle_\pi = \sum_{n=0}^{N} \langle j, K^n k\rangle_\pi + \langle j, K^{N+1} Rk\rangle_\pi.$$

Thus a Green–Kubo coefficient is a correlation sum plus an exact displayed remainder, rather than a silently assumed infinite series.

For the full-fibre conditional-resampling kernel P_C, idempotence gives $P_C^n = P_C$ for every $n \geq 1$. Positive-lag correlations are therefore constant. A fibre-centered current with $P_C j = 0$ has no positive-lag memory, while a nonzero one-step correlation makes consecutive partial sums differ by the same nonzero amount. One full-fibre repair projector cannot be a decaying, nonzero-memory transport dynamics.

On a finite closed oriented graph, typed positive cell volumes and heat capacities convert concentration to particle amount and temperature to energy. Fick and Fourier fluxes then define canonical one-step updates whose total changes equal the clock increment times the total source and hence vanish when the source sums vanish. Nonnegative conductance makes the flux–gradient pairing nonpositive; explicit negative-conductance controls reverse that sign.

Proof. Detailed balance makes L self-adjoint, and the exact Dirichlet identity expresses $\langle f, Lf\rangle_\pi$ as a nonnegative weighted sum of squared differences. Applying this to linear combinations of the Poisson solutions proves symmetry and the full finite quadratic-form inequality. The cutoff formula telescopes the Poisson equation. Projector idempotence gives the positive-lag dichotomy. Graph summation by parts and cancellation of internal oriented edges give the constitutive and global-balance statements. Every step, including two-state and two-vertex witnesses and negative controls, is checked in Lean [7]. □

The theorem does not select a physical transition generator, distance, clock, boundary condition, conductance, or readout. It proves no stability, hydrodynamic limit, or measured transport coefficient. In particular, the projector obstruction requires a source-derived nonidempotent local or random-scan effective evolution before OPH can make a long-memory transport claim.

6.1 Protected records with decaying repair memory

A conditional expectation describing equilibrium need not be the transition operator. This distinction admits conserved records and dissipative memory on one finite state space. The following construction supplies a local stochastic law and its faithful reference; neither is identified with a particular physical source.

Let $X = \coprod_c (\{c\} \times U_c \times V_c)$ be finite, with nonempty coordinate sets and a strictly positive probability law π. On the real space $L^2(\pi)$, let P be conditional expectation given c, E_u conditional expectation given (c, v), and E_v conditional expectation given (c, u). Their stochastic interpretations resample only the omitted coordinate from the same reference. Define

$$T = \tfrac{1}{2} I + \tfrac{1}{4} E_u + \tfrac{1}{4} E_v. \tag{3}$$

Thus bounded patches carry local state (c, u, v), a protected record c, coordinate readback ports and conditional repair moves. The step index is a declared update count, with no laboratory time calibration.

Proposition 6.2 (A protected-record dynamics with finite correlation tails)**.** *The kernel T is stochastic, reversible with respect to π, and fixes every function of c. It satisfies $TP = PT = P$. There is $0 < \varepsilon < 1$ such that the entrywise inequality $T^2 \geq \varepsilon P$ holds. Consequently, on* $\ker P$*,*

$$\|T^n\| \leq (1-\varepsilon)^{\lfloor n/2 \rfloor}, \qquad \operatorname{gap}(I-T) \geq \varepsilon/2. \tag{4}$$

The unique fibre-centred Poisson solver is $R = (I - T + P)^{-1} - P$. For $Pf = Pg = 0$,

$$\mathcal{G}(f,g) = \langle f, Rg \rangle_\pi = \sum_{n=0}^{\infty} \langle f, T^n g \rangle_\pi \tag{5}$$

is symmetric and positive semidefinite. After retaining lags 0 through N, its exact remainder and a uniform bound are

$$\begin{aligned} \mathcal{G}(f,g) - \sum_{n=0}^{N} \langle f, T^n g \rangle_\pi &= \langle f, T^{N+1} R g \rangle_\pi, \\ \left| \mathcal{G}(f,g) - \sum_{n=0}^{N} \langle f, T^n g \rangle_\pi \right| &\leq \frac{\|f\|_\pi^2 + \|g\|_\pi^2}{\varepsilon} (1-\varepsilon)^{\lfloor (N+1)/2 \rfloor}. \end{aligned} \tag{6}$$

Proof. Conditional expectations are orthogonal projections in $L^2(\pi)$; their kernels satisfy detailed balance and preserve c. The tower property gives $TP = PT = P$, and $T \geq I/2$ as an operator. Two consecutive repairs, first one coordinate and then the other, can reach every state in the same fibre with positive probability. Finiteness and faithfulness therefore give the displayed minorization. The remainder $Q = (T^2 - \varepsilon P)/(1-\varepsilon)$ is stochastic and stationary for π. Jensen's inequality gives $\|Q\| \leq 1$. On $\ker P$ this implies (4), using $1 - \sqrt{1-\varepsilon} \geq \varepsilon/2$ for the gap. The Neumann series on that subspace defines R, while R vanishes on $\operatorname{ran} P$. Reversibility makes R self-adjoint, and its eigenvalues there are $(1-\lambda)^{-1} > 0$. The finite geometric-series identity gives the exact remainder. Sum $|\langle f, T^n g \rangle| \leq \|f\|\|g\|(1-\varepsilon)^{\lfloor n/2 \rfloor}$ from $N+1$ onwards, grouping successive pairs and using $2\|f\|\|g\| \leq \|f\|^2 + \|g\|^2$. □

An exact eight-state instance takes $c, u, v \in \{0,1\}$, fibre masses $(1/3, 2/3)$, and conditional reference weights $(1,2,3,4)/10$ and $(4,1,2,3)/10$ in lexicographic (u,v) order. Both references are nonproduct laws. Its sharp two-step minorization is $\varepsilon = 7/96$, giving a certified gap lower bound $7/192$. For $f = u - Pu$ and $g = v - Pv$, the one-sided coefficient matrix is exactly

$$\begin{pmatrix} 11417/9375 & 1832/3125 \\ 1832/3125 & 3816/3125 \end{pmatrix},$$

and the positive-lag autocorrelation $\langle f, Tf \rangle = 443/2400$ is nonzero. An independent rational verifier reconstructs the coordinate kernels, checks both Poisson inverse identities, and replays 65 correlation lags, the exact cutoff remainder and eight distribution readouts.

Fibre centering is essential: $c - 2/3$ has global mean zero but is fixed by T, with persistent correlation $2/9$ and no centred Poisson solution. The limiting distribution retains its initial fibre masses. This is not global ergodicity, an identification of T with the full-fibre projector, or a dynamic intertwiner recovering a recurrent mode from that projector. It leaves the obstruction for the distinct source-counted transition artifact intact. The local stochastic law and reference are supplied; physical source attachment, transport normalization, clock calibration and a continuum limit require additional arguments.

Maintained records in a noisy coherent source. A declared coherent-code source realizes bounded self-reading patches with local state, ports, readback, noisy feedback and actual public record banks. Complete transfer/reset services reduce carrier leakage, and a protected quantum computation exports into a maintained classical archive. With a fixed positive local service error below a derived library bound, $O(\log\log q)$ quantum protection levels, staged decoding and expander refresh of $O(\log q)$-bit words make the joint decoded quantum and live record error tend to zero. The archive remains readable throughout each refresh; no perfect bare decision bit or fault-free runtime controller is assumed. The analytical argument imports computation-code fault tolerance and an explicit expander bound; finite tests check its interfaces and fault controls, without executing the enormous all-size apparatus. Coherent code gluing, the local noise law, external schedule and cofinal control/inventory resources are hypotheses. They are not consequences of classical record agreement. Logical accounting, physical energy, calibrated time and source-selected matter dynamics are distinct.

For a separate deterministically prepared filled vacuum, the same massive walk has a positive normal-ordered excitation Hamiltonian with exactly its integer-time Fock channels. Both carrier bands enter the local covariance limit. Complete resolved-mode number reads retain the occupied background and have bounded mean excitation cost in the stated smooth carrier class; a finite analytical clock witness has probability error below 0.038 and swing above 0.923. Its enormous three-dimensional apparatus is not simulated. The longer sea preparation has its own charged protection and storage bounds, so the empty-vacuum resource estimates do not apply to it. High-band Floquet shifts preserve these integer-time reads and infrared limits while changing ultraviolet excitation costs. This positive stroboscopic generator identifies neither laboratory energy nor the work of the supplied controls.

A native stationary history source. Fair integer seam repair continued within a balanced fixed-total sector supplies a concrete nonidempotent evolution. On a connected weighted graph of p ports with $0 < r < p$ raised loads, it is the symmetric exclusion chain with generator $Gz = -Lz/2$ for $z = n - (r/p)\mathbf{1}$. Under the uniform stationary law, $\mathrm{Cov}(z) = \kappa\Pi$, where $\kappa = r(p-r)/(p(p-1))$ and $\Pi = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/p$. Consequently a fixed linear local readout B has history limit

$$T^{-1/2}\int_0^T Bz(t)\,dt \Longrightarrow \mathcal{N}(0, 4\kappa BL^{+}B^{\mathsf{T}}).$$

This is a fixed-graph additive-functional limit for independent Poisson attempt clocks. Its ordinary time average has mode variance $\kappa h(\lambda T/2)$, with $h(u) = 2(u-1+e^{-u})/u^2$, rather than a freely normalized Gaussian source. The screen-microphysics paper gives the martingale proof, the nonlinear readout decomposition and the correction for pre-attempt incident-event records. Exact finite-state witnesses and independent numerical controls are retained in the public repository [7]. Occupancy, record gain, elapsed-time convention and the physical meaning of the readout require specification. The result supplies neither a uniform refinement limit nor a measured primordial amplitude or tilt.

The same package carries the entropy-balance shape used by the gravitational branch. For any state and faithful reference the exact identity $S(p) - S(\tau) = \langle K\rangle_p - \langle K\rangle_\tau - D(p\|\tau)$ holds, so the entanglement first law $\delta S = \delta\langle K\rangle$ is this identity with the quadratic-order deficit dropped; through the central split $K = 2\pi B + Z$ it reads

$$\Delta S = 2\pi\Delta\langle B\rangle + \Delta\langle Z\rangle - \Delta D(\,\cdot\,\|\tau),$$

where $\Delta D = D(p'\|\tau) - D(p\|\tau)$. For the reference-preserving repair step, data processing gives $\Delta D \le 0$, and one repair step with a central charge measurable through the repaired visible da-

tum obeys the cap Clausius inequality $2\pi\Delta\langle B\rangle \le \Delta S$, since the kernel also conserves every fibre-measurable charge and hence $\Delta\langle Z\rangle = 0$. The finite first-law premise package of the Einstein branch is discharged on this model: on the tangent space of the probability simplex the entropy differential at the reference equals the modular pairing as a theorem, the modular pairing distributes through the central split, and the central pairing is the edge differential, so the branch's finite bulk-edge-central first law applies with no assumed premise beyond the split of the reference. Repair displacements are mass preserving with vanishing central pairing, so the first-order entropy change of one repair step is purely bulk-modular. The physical attachment of the split stays with the energy-clock receipt.

The Lean development carries the exact finite statements: relative entropy nonnegativity, the log-sum inequality, data processing, the conditional-resampling optimizer and its kernel package, the Gibbs Pythagorean identity and minimizer, the first-law split, the low-temperature bound, the fluctuation identities with reciprocity, the finite Green–Kubo matrix and exact cutoff remainder, the repair-projector memory obstruction, the typed Fick and Fourier updates, the exact-remainder cap first law, the discharged Einstein-branch first-law premises, and the Kraus completeness of partition pinching; the modules are `FiniteConditionalRepair`, `FirstLawIdentity`, `FluctuationTheorems`, `GreenKubo`, `GraphDiffusion`, `CapFirstLaw`, `EinsteinPremiseLink`, `StationaryRealization`, and `PartitionPinchingCP` in the Lean thermodynamics and event-algebra libraries, and the exact-rational certificate replays the kernel algebra and fluctuation identities and records the entropy checks [7]. The source-realization interface needs less than reversibility: every stochastic kernel that preserves a faithful reference contracts relative entropy to that reference. Detailed balance is not a premise. The exact lazy directed three-cycle is stationary and violates detailed balance, yet satisfies this second law; the Lean module `StationaryRealization` proves the separation. Detailed fluctuation relations and Onsager reciprocity therefore remain additional receipts rather than consequences of entropy descent.

The pinned source-counted collar table has also been exhausted within its declared coordinate-projection grammar. All $2^4 - 1 = 15$ syntactic maps obtained by retaining a nonempty subset of its four committed packet fields were audited on 1028 observers and 31744 counted transitions, without an additional observer-patch run. Because the record-family and S_3-sector fields are constant on this run, those maps induce only four distinct partitions. The repair-load count aggregation is an eight-state stochastic kernel that is irreducible and aperiodic and has a full-support stationary law. It is not a certified Markov quotient of the fine chain: its strong-lumpability defect, the largest outgoing target-block mass difference between two fine states in one repair-load block, is 0.911587983 on the pinned floating-point table. On that table its detailed-balance defect is $2.009924089 \times 10^{-4}$; from the uniform initial law the sampled relative entropy falls from 5.00947 to 1.84009 over sixteen steps. This is a nonreversible finite H-theorem probe, not a microscopic-reversibility receipt. It does not close the realization: the only observed record-family value makes protected-charge preservation vacuous, and the stationary law is not identified with the state optimizer's source reference. On the fine twenty-state quotient the only closed communicating class is the singleton freezeout state, so the alternative recurrent-class restriction is a trivial equilibrium rather than a nontrivial relaxation model. These results exhaust only the declared coordinate maps and the fine-chain recurrent restriction. They do not enumerate arbitrary partitions or nonlinear, statistical, stochastic, weakly lumpable, or history-dependent quotient maps of the retained artifact. The audit excludes none of those routes, nor changed source dynamics or a bounded run with a nonconstant protected record.

The state table/reference was locally pre-specified and hash-pinned; the recurrent transition chain was extracted post hoc from retained run data. This does not repair the failure. The state-side conditional-resampling action is idempotent, whereas the exact two-state recurrent transition

action has the nonconstant eigenpair

$$Pv = \frac{665437}{726948}v, \qquad v = (54356, -7155), \qquad 0 < \frac{665437}{726948} < 1.$$

If an idempotent action H and this transition action P were related by $HT = TP$, applying idempotence twice gives $\lambda(\lambda - 1)Tv = 0$, hence $Tv = 0$. Thus every such intertwiner erases the exact extracted mixing mode. The transition stationary mass 7155/61511 also lies strictly between the adjacent empirical masses 1905/16384 and 1906/16384; no deterministic coarse graining of the 16384-sample state reference produces it. A freely chosen stochastic coupling could evade the denominator test, but would be a new premise rather than source evidence. These exact obstructions exclude the two audited direct bindings (a mixing-mode-preserving linear intertwiner into the idempotent heat bath and a deterministic empirical pushforward) on the committed artifact. They do not exclude stochastic, nonlinear, reverse-direction, or enriched-source constructions. The committed random-scan preflight decides the random-scan instance of the stochastic route within its certified grammar: on every computed subset (the eleven subsets of at least two of the four committed packet fields on the 26-state alphabet, under the visit-count reference, and the record/companion pair on the realized 256-state structure, under the occupation reference), the random-scan mixture of conditional-resampling projectors under both declared schedulers is row-stochastic and exactly stationary for the one shared reference of its arena. The committed uniform-scheduler mixtures are additionally certified non-idempotent, so they escape the idempotent spectral obstruction; the exact fixed space of every computed mixture under either scheduler is one-dimensional, so the only observables it protects are constants. The remaining step-field subsets contain a certified-constant field; their mixtures carry the strictly positive full-space resampler as a component, which forces the same one-dimensional fixed space without any claim of non-idempotence. The only retained step field beyond the packet fields is certified constant on the run, so the retained step record admits no enrichment; the retained non-step cell fields are outside the certified grammar. The recorded continuations are a fresh export of a dynamically conserved label whose join with a committed field disconnects, or a dilated construction outside the random-scan grammar.

The conditional package retains five typed requirements: the global representation of the weighted local objective for both optimizer instantiations; the common source-derived reference shared by state and transition optimizers, which forbids manufacturing the transition matrix from a desired equilibrium output; the identification of the actual source-derived collar transition matrix with the conditional-resampling kernel through the existing matrix receipt; stochasticity, stationarity, and protected-charge preservation alone certify only the weaker stationary H-theorem branch and do not discharge that full premise-A3 receipt, while detailed balance is additionally required wherever microscopic reversibility is claimed; the identification of one modular charge with physical energy and a calibrated clock, of which the finite central-interface split $K_C = 2\pi B_C + Z_C$ is the attained cap-branch part; and uniform low-temperature spectral-tail control on a genuinely cofinal family for a continuum third law. The committed cofinal interface provides the ambient order, cofinal embedding, no-terminal-stage theorem, strict carrier growth, and same-family concentration theorem, while supplying no source or physical continuum interpretation. Thus a source-produced family inhabiting that interface and its physical regulator attachment are unsupplied. The finite gap-plus-cardinality theorem is one sufficient route rather than a necessary condition. Every proof in the package is short and elementary once the framework supplies its two inputs, the repair fibre as the transition feasible set and the reference shared with the state optimizer. The four-law identities then require no additional thermodynamic premise or deep theorem; the work lies in the decomposition, which the certified strict-descent counterexample shows is necessary. On a source object discharging the named receipts, ordinary finite thermodynamic identities follow from the

instantiated observer model rather than being postulated separately.

6.2 Elementary corollaries of the record and screen structure

The finite record algebra, the celestial sphere and the information projection carry several textbook laws as direct corollaries. Each is stated below with the exact identity it rests on and with the identification it does not supply. Exact rational controls for every statement, pinned to this section, are part of the repository receipts [7].

Repeated reads and the Zeno limit.

Proposition 6.3 (Repeatability and the Zeno limit)**.** *Let ρ be a state on the finite accessible algebra of one completed slice, P an event projector with $w = \mathrm{Tr}(\rho P) > 0$, and $\rho_P = P\rho P/w$ the declared Lüders update. Then $\mathrm{Tr}(\rho_P P) = 1$, and the update is idempotent. Let H be a Hermitian generator on the same space, $U_s = e^{-isH}$, and let the observer reread P after each of n equal steps of total parameter t. For ρ supported in the range of P, the probability p_n that all n rereads return P satisfies*

$$1 - p_n \le \frac{2t^2\|H\|^2}{n},$$

so $p_n \to 1$ as $n \to \infty$, and the conditioned state converges to $e^{-itPHP}\rho\, e^{itPHP}$.

Proof. Repeatability and idempotence are the Lüders theorems of the finite event-algebra library [6, 7]. Put $s = t/n$, $A = PU_sP$ and $B = Pe^{-isPHP}P$; both are contractions on the range of P. For Hermitian X the exponential series gives $\|e^{-isX} - (1 - isX)\| \le s^2\|X\|^2/2$, hence $\|A - B\| \le s^2\|H\|^2/2 + s^2\|PHP\|^2/2 \le s^2\|H\|^2$. Telescoping, $\|A^n - B^n\| \le n\|A - B\| \le t^2\|H\|^2/n$. The joint probability of n returns is $p_n = \mathrm{Tr}(A^n\rho A^{n\dagger})$, while $B^n = Pe^{-itPHP}P$ is unitary on the range of P, so $\mathrm{Tr}(B^n\rho B^{n\dagger}) = 1$. For contractions $|\,\mathrm{Tr}(X\rho X^\dagger) - \mathrm{Tr}(Y\rho Y^\dagger)| \le 2\|X - Y\|$, which gives the bound; the convergence of the conditioned state follows from the same estimate. □

A rational control replaces the exponential by the rotation with $\tan(\theta_n/2) = 1/n$, so $\cos\theta_n = (n^2 - 1)/(n^2 + 1)$, read through a rank-one projector after every step. The survival with reads is $\cos^{2n}\theta_n$, which obeys $1 - \cos^{2n}\theta_n \le 4/n$, while the survival without reads is $\cos^2(n\theta_n)$ and stays away from one. The step parameter is a flow parameter and H is supplied; neither is a physical clock or a source-selected generator.

Distinguishable records through a port.

Proposition 6.4 (Port capacity)**.** *Let the interface algebra of port p act on a Hilbert space of dimension d_p. A family of states that one read through the port distinguishes without error has at most d_p members. One read therefore carries at most $\log_2 d_p$ bits, and one read of all twelve ports of a carrier at most $\sum_p \log_2 d_p$ bits.*

Proof. States distinguished without error by one measurement have pairwise orthogonal supports, and pairwise orthogonal nonzero vectors are linearly independent, so there are at most d_p of them. The port bound adds over the ports because the boundary space is their tensor product. □

Holevo's bound, an imported result, gives the same ceiling for reads with error. Neither statement assigns an energy to a record: the energy of one distinction enters only through the clock and energy calibration, and the bound itself is a count.

Celestial kinematics.

Proposition 6.5 (Aberration, the null scale factor and rapidity addition)**.** *Let the celestial sphere carry the action of $SL(2,\mathbb{C})$ by Möbius maps of the spacetime and Einstein derivation [3], with stereographic coordinate ζ, and let each null direction carry the null vector $q = (1, \Omega)$. Realize the Lorentz group on Hermitian 2×2 matrices by $X \mapsto AXA^\dagger$, under which q is the rank-one matrix $\kappa\kappa^\dagger$ with $\kappa = (1, \zeta)$. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in SL(2,\mathbb{C})$:*

1. *the direction transforms as $\zeta \mapsto (c + d\zeta)/(a + b\zeta)$;*
2. *the time component of q is multiplied by $\lambda_A(\zeta) = (|a + b\zeta|^2 + |c + d\zeta|^2)/(1 + |\zeta|^2)$, and $\lambda_A \equiv 1$ for every $A \in SU(2)$;*
3. *for $A_k = \mathrm{diag}(k^{-1/2}, k^{1/2})$ with $k > 0$, put $v = (k - 1/k)/(k + 1/k)$, $\gamma = (k + 1/k)/2$ and $\cos\theta = (1 - |\zeta|^2)/(1 + |\zeta|^2)$. Then $\lambda_{A_k} = \gamma(1 - v\cos\theta)$ and the transformed direction obeys $\cos\theta' = (\cos\theta - v)/(1 - v\cos\theta)$;*
4. *$A_{k_1} A_{k_2} = A_{k_1 k_2}$, so $v(k_1 k_2) = (v_1 + v_2)/(1 + v_1 v_2)$.*

Proof. With $X = \kappa\kappa^\dagger$ the time component is $(|\kappa_0|^2 + |\kappa_1|^2)/2$ and the axial component is $(|\kappa_0|^2 - |\kappa_1|^2)/2$, which gives $\cos\theta$. Under A, $\kappa \mapsto A\kappa = (a + b\zeta, c + d\zeta)$, which is (1) and (2); $SU(2)$ preserves $|\kappa|$. For A_k, $A\kappa = (k^{-1/2}, k^{1/2}\zeta)$, so $\lambda = (1/k + k|\zeta|^2)/(1 + |\zeta|^2)$ and $\zeta' = k\zeta$; substituting $|\zeta|^2 = (1 - \cos\theta)/(1 + \cos\theta)$ gives (3). Diagonal matrices multiply entrywise, which is (4). □

Read with θ the angle between the propagation direction of a null ray and the axis, and v the axial velocity of a second frame, (3) is the aberration law and λ is the Doppler factor of that frame; (4) is the relativistic addition of collinear velocities. The scale factor is a statement about the null vector on the declared sphere. It becomes a frequency ratio only through the clock calibration and the physical spacetime attachment, which this section does not supply.

The canonical distribution as the information projection.

Proposition 6.6 (Gibbs law from one linear constraint)**.** *Let τ be a faithful reference law on a finite record set, E a real function on it, and $g_\beta = \tau e^{-\beta E}/Z_\beta$. For every law p with $\langle E\rangle_p = \langle E\rangle_{g_\beta}$,*

$$D(p\|\tau) = D(p\|g_\beta) + D(g_\beta\|\tau).$$

Hence g_β is the unique minimizer of $D(\cdot\|\tau)$ among laws with that expected value: the information projection of the third axiom under one linear constraint is the canonical distribution, and β is the multiplier of the constraint.

Proof. $\log(g_\beta/\tau) = -\beta E - \log Z_\beta$ is affine in E, so $D(p\|\tau) - D(p\|g_\beta) = \sum_x p(x)\log(g_\beta/\tau)(x) = -\beta\langle E\rangle_p - \log Z_\beta$, which by the constraint equals the same expression under g_β, namely $D(g_\beta\|\tau)$. Uniqueness follows from $D(p\|g_\beta) = 0$ only at $p = g_\beta$. □

The history-level form of this identity and the exponential form of the projected path law are machine-checked in the information-projection library [7]. The energy ledger E is declared, and β is free against the modular Hamiltonian alone; a temperature reading consumes the calibration row.

Occupation laws under a declared exchange sector.

Proposition 6.7 (Fermi–Dirac, Bose–Einstein and truncated occupation)**.** *Consider one mode of energy ε at chemical potential μ and inverse temperature β, and put $x = e^{-\beta(\varepsilon-\mu)}$. The Gibbs law of Proposition 6.6 on the occupation ledger, with weight x^n for occupation n, has mean occupation $x/(1+x) = 1/(e^{\beta(\varepsilon-\mu)}+1)$ when the mode admits the occupations $\{0,1\}$, $x/(1-x) = 1/(e^{\beta(\varepsilon-\mu)}-1)$ when it admits every $n \geq 0$ and $x < 1$, and $\sum_{n\leq m} nx^n / \sum_{n\leq m} x^n$ when it is cut at m; the last equals the first at $m = 1$ and increases to the second as $m \to \infty$.*

Proof. Geometric sums: $\sum_{n\geq 0} x^n = 1/(1-x)$ and $\sum_{n\geq 0} nx^n = x/(1-x)^2$; the two-term case and the truncated case are the same sums cut off. Monotonicity in m holds because each added term has occupation above the previous mean. □

The exchange sector is a declared premise of this proposition. Nothing in it selects the sector, and the two laws stand as the two admissible sectors of one information projection.

Coarse-graining of the record law.

Proposition 6.8 (Record identification does not raise the entropy of the record law)**.** *Let $r\colon \Sigma \to \Sigma'$ be a map on public records that identifies some of them, and let p be a law on Σ with push-forward r_*p. Then $H(r_*p) \leq H(p)$ for the Shannon entropy and $\sum_y (r_*p)(y)^2 \geq \sum_x p(x)^2$, with equality in both exactly when r is injective on the support of p. If K is a stochastic repair kernel with stationary law π, then $D_f(pK\|\pi) \leq D_f(p\|\pi)$ for every f-divergence, in particular for the relative entropy and the χ^2 divergence.*

Proof. $H(r_*p) = H(p) - H(p \mid r_*p)$ and the conditional entropy is nonnegative, vanishing exactly when r is injective on the support. For the collision sum, $(a+b)^2 \geq a^2+b^2$ for $a, b \geq 0$, with equality only when one vanishes. The kernel statement is the data-processing inequality for f-divergences applied to the pair (p, π) with $\pi K = \pi$. □

This is a statement about the law of the public records under repair. It does not identify that entropy with a thermodynamic entropy; the four-law package makes that identification under its declared repair law and calibration.

None of the six statements supplies a physical clock, energy, frequency, exchange sector or source selection. Their standing in the observation ledger is recorded row by row with the premises each consumes.

7 Closure read by the same observers

Closure requires a candidate history to reproduce the specification that its observers read. Choosing a carrier, scheduler or memory law is permitted; the complete choice and calibration ancestry must be explicit. A fixed point of a chosen numerical map is not by itself an identification of that map with a physical readout.

7.1 Pixel ratio, currents and mass response

Let $P = a_{\mathrm{cell}}/\ell_\star^2$ and $\varphi = (1+\sqrt{5})/2$. The conditional self-similar entropy balance $S_{\mathrm{gen}}/S_{\mathrm{bulk}} = S_{\mathrm{bulk}}/\langle L\rangle$ gives $x^2 - x - 1 = 0$, hence $x = \varphi$. The identification of its detuning with an electromagnetic reading is a separate same-quantity premise:

$$\alpha_{\mathrm{ext}}(P) = \frac{P-\varphi}{\sqrt{\pi}}, \qquad A_T(P) = \alpha_{\mathrm{Th}}^{-1}(P), \qquad P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}.$$

Here A_T must be computed from the same field theory and charged spectrum whose records implement P. If the displayed map preserves a closed interval and has derivative bounded by $q < 1$, Banach contraction supplies one fixed point there. Certified roots for specified maps prove that mathematical statement; the physical Thomson transport and map selection remain independent obligations.

The current finite construction makes the connection operational. The complete-response gauge theorem restricts the current algebra under its faithfulness, commutator-closure and inner-transport hypotheses [1]. A declared memory law produces a P-dependent flavor coefficient; the radial localized CAR inclusion intertwines that coefficient and the actual current matrices on one carrier [2]. Consequently a candidate screening calculation must use these charged states and their actual Hamiltonian, including any interaction and spatial band shift. A compact opposite-degree defect pair has $A \neq 0$, while the exact radial quartic construction uses a zero-mode band. They cannot share the quadratic feedback formula without an additional derivation.

One explicit feedback family has

$$\frac{M_U}{E_\star} = e^{-2\pi} P^{1/6}, \qquad \frac{v}{E_\star} = P^{-1/2} e^{-2\pi/(\beta a)}, \qquad \alpha_i^{-1}(\mu) = a^{-1} + \frac{b_i}{2\pi} \log \frac{M_U}{\mu},$$

with declared $b = (41/10, -19/6, -7)$ and an edge heat-sum equation for a. The heat sums are infinite sums with certified tails. The coefficient β is a design choice, not the weak beta-function coefficient. For $\beta = 4$, $P \geq 1$, $M_U \geq m_Z$, positive couplings and no thresholds, the full feedback equations have no solution. The alternative specified $\beta = 7$ family admits the enclosed conditional root

$$P = 1.628533369611833\ldots, \qquad A = 168.815082928453\ldots.$$

This closes that finite design's equations. Its intended chiral running census, missing doubler-removal attachment, physical endpoint transport and interaction-dependent response prevent interpreting it as the observed fine-structure constant or a complete particle spectrum. Exact equations, interval bounds and alternative feedback domains are in the particle paper and `model/pixel_mass_closure/`.

7.2 Physical scale and calibration ancestry

If a dimensionless atomic calculation supplies the isolated cesium gap $\varepsilon_{\mathrm{Cs}} = \hbar\omega_{\mathrm{Cs}}/E_\star$ with $E_\star = \hbar c/\ell_\star$, then

$$\ell_\star = \frac{c\varepsilon_{\mathrm{Cs}}}{2\pi\nu_{\mathrm{Cs}}}, \qquad G_{\mathrm{SI}} = \frac{c^5\varepsilon_{\mathrm{Cs}}^2}{4\pi^2\hbar\nu_{\mathrm{Cs}}^2}.$$

A prediction of G by this route requires source-derived atomic, nuclear, QCD, electromagnetic and mass data with no measured-G ancestor. Substituting a gravity-calibrated gap is instead a change of units. Similarly, assigning SI c fixes a length/time ratio and does not determine all particle masses.

7.3 Global closure for total capacity

The global counterpart of the local P-closure begins with a supplied positive integer capacity-carrier dimension

$$D = \dim \mathcal{H}_{\mathrm{cap},r,D}, \qquad N = \log D.$$

The universe-level readback equation is

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

Here the argument fixes the type. In every occurrence, M_0 is a multiplicative code size. Once the complete terminal fiber scalarizes, the universe-level notation means

$$M_0(\mathfrak{U}_N) := \widehat{F}_{r,0}(e^N).$$

Thus N is the logarithmic capacity read back from the trial universe, while $M_0(\mathfrak{U}_N)$ is the number of correctable public records. The equation is the logarithmic form of stable direct capacity closure, not a new producer or an appeal to a measured target. The carrier type is frozen in the branch contract. Boundary, total, edge-center, and other Hilbert spaces are not interchangeable. Let $\widetilde{\Omega}_{r,D}$ be the terminal physical quotient fiber reached from the source-derived trial universe $\mathfrak{U}_{r,D}$. Its members are repair-normal and pass the declared local, obstruction, and provenance gates. Membership contains no capacity-equals-D predicate.

For $q \in \widetilde{\Omega}_{r,D}$, observer O has a finite commutative record algebra $\mathcal{R}_O(q)$ with atom set $X_O(q) = \operatorname{At} \mathcal{R}_O(q)$. Each shared interface e has an atom set $X_e(q)$ and a source-derived record-atom readout

$$r_{Oe} : X_O(q) \longrightarrow X_e(q).$$

This atom-level construction is used because the generic visible maps on full observer algebras need not be unital *-homomorphisms in the direction required by an algebra equalizer. The compatible public global sections are

$$X_{\mathrm{pub}}(q) = \{(x_O)_O : r_{Oe}(x_O) = r_{O'e}(x_{O'}) \text{ on every shared interface}\}, \qquad \mathcal{R}_{\mathrm{pub}}(q) = C(X_{\mathrm{pub}}(q)).$$

Theorem 7.1 (Public global sections)**.** *For every finite observer record diagram, $X_{\mathrm{pub}}(q)$ is finite and $C(X_{\mathrm{pub}}(q))$ is a finite commutative unital C^*-algebra. Every isomorphism of observer record diagrams induces a bijection of public global sections and a *-isomorphism of their function algebras.*

Proof. The public sections form the subset of the finite product $\prod_O X_O(q)$ satisfying finitely many equalities. A diagram isomorphism transports the atom sets and interface maps, hence compatible tuples bijectively; pullback transports their function algebras. □

Only endogenously writable records count. Define

$$X_{\mathrm{reach}}(q) = \{x \in X_{\mathrm{pub}}(q) : x \text{ terminates an admissible endogenous semantic history}\}.$$

The history may use declared source data, accepted repair, semantic continuation, and quotient-visible writes. It may not use a target record, supplied-capacity metadata, measured Λ, or the electroweak bridge. The publicness policy is a frozen nonempty family

$$\mathfrak{P}(q) \subseteq 2^{\mathcal{O}(q)} \setminus \{\varnothing\}$$

of authorized observer subsets. Collective, universal-local, and quorum publicness are distinct theorem branches.

For every $x \in X_{\mathrm{reach}}(q)$, authorized set $A \in \mathfrak{P}(q)$, and allowed same-interface semantic continuation κ, the source construction must emit a joint stochastic kernel

$$K_{A,\kappa}(y \mid x), \qquad y \in Y_{A,\kappa},$$

whose observer marginals agree with the local checkpoint packets. The joint kernel is an independent receipt. It cannot be manufactured from local marginals.

Theorem 7.2 (Local marginals do not determine public capacity)**.** *There are two binary-input joint checkpoint channels with identical observer-by-observer marginals and different public capacities.*

Proof. Let U be uniform. The channel $(Y_1, Y_2) = (U, U \oplus x)$ has uniform input-independent marginals, while the joint parity recovers x. Independent uniform outputs have the same marginals and carry no information about x. Thus the joint capacities are two and one, respectively. □

For a channel K, write $S_K(x) = \{y : K(y \mid x) > 0\}$. A code $C \subseteq X_{\mathrm{reach}}(q)$ is zero-error correctable for K when a decoder d_K satisfies

$$K(d_K^{-1}(x) \mid x) = 1 \qquad (x \in C).$$

Theorem 7.3 (Support criterion and compound zero-error capacity)**.** *A code is zero-error correctable for K exactly when the supports $S_K(x)$, $x \in C$, are pairwise disjoint. Let $\mathfrak{K}(q)$ contain every authorized checkpoint channel and every declared finite composition. Define the compound confusability graph G_q on $X_{\mathrm{reach}}(q)$ by joining $x \neq x'$ whenever their output supports overlap for at least one $K \in \mathfrak{K}(q)$. Then the exact public-record capacity is*

$$\boxed{M_0(q) = \alpha(G_q)},$$

where α is the graph independence number.

Proof. A successful decoder cannot assign one output to two inputs, which proves the support criterion in one direction. Pairwise disjoint supports admit the decoder that returns the unique supporting input. A code valid for every declared channel is therefore exactly an independent set in the union of the channel confusability graphs. □

For exact indefinite continuation, replace each kernel by its support relation $R_K = \{(x, y) : K(y \mid x) > 0\}$. Boolean relation composition generates at most $2^{|X_{\mathrm{reach}}|^2}$ relations, so the support semigroup and its compound confusability graph close after finitely many steps.

Theorem 7.4 (Reversible fast branch)**.** *If every authorized continuation is a known injective deterministic map on $X_{\mathrm{reach}}(q)$, then*

$$M_0(q) = |X_{\mathrm{reach}}(q)|.$$

Proof. Injectivity keeps every pair of inputs distinguishable for every continuation, so G_q has no edges. □

Checkpoint invariance is not the capacity definition. A cyclic permutation of m labels has a one-dimensional fixed function algebra, while its inverse decodes all m labels. Thus a fixed-projector count can undercount the correctable capacity arbitrarily.

For a frozen finite channel family or continuation horizon and tolerance ε, let $M_\varepsilon(q)$ be the largest code for which every declared channel has a decoder with worst-input error at most ε.

Theorem 7.5 (Approximate-capacity stability and zero-error discontinuity)**.** *If corresponding channel rows obey*

$$\sup_{K,x} d_{\mathrm{TV}}(K(\cdot \mid x), \widehat{K}(\cdot \mid x)) \leq \delta,$$

then

$$M_{\varepsilon+\delta}(\widehat{\mathfrak{K}}) \geq M_\varepsilon(\mathfrak{K}).$$

Zero-error capacity is discontinuous at every identity channel on more than one symbol: adding arbitrarily small full-support noise collapses M_0 to one.

Proof. Total-variation control changes the probability of each correct-decoding event by at most δ. For the second statement, $(1-\delta)I+\delta U$, with U full support, makes every pair of inputs confusable for every $\delta > 0$. □

The capacity carrier is an exact finite Hilbert space $\mathcal{H}_{\mathrm{cap},r,D}$ of dimension D. Each reachable public record class has a nonzero projection P_x on this carrier, and distinct records have orthogonal projections.

Theorem 7.6 (Carrier bound and saturation rigidity)**.** *Every correctable public code C satisfies*

$$|C| \le D, \qquad \log M_\varepsilon(q) \le \log D = N.$$

If $M_0(q) = D$, the code projections are rank one and sum to the identity.

Proof. The ranks of nonzero orthogonal projections add and their sum has rank at most D. Equality with D codewords forces every rank to be one and the sum to have full rank. □

The primary finite readback remains set-valued:

$$\boxed{\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.}$$

A scalar $\widehat{F}_{r,\varepsilon}(D)$ exists exactly when the nonempty terminal fiber has one common defined value. The evaluator must distinguish an empty readback, an ambiguous readback, and a singleton scalar readback. Existential closure means $D \in \mathfrak{F}_{r,0}(D)$; selector-free stable closure means

$$\boxed{\mathfrak{F}_{r,0}(D) = \{D\}.}$$

Equivalently, the unclosed kernel

$$K_r(D,m) = |\{q \in \widetilde{\Omega}_{r,D} : M_0(q) = m\}|$$

has support only at $m = D$. One saturated branch is insufficient when the terminal fiber is ambiguous.

The full capacity packet consists of observer lineage, record atoms, interface maps, reachable public sections, publicness policy, global kernels, and carrier projections. An isomorphism preserving that packet transports every exact confusability edge, approximate decoder error, and projection rank. Hence M_ε is implementation invariant under packet isomorphism.

Once scalarized, the active map $f(D) = \widehat{F}_{r,0}(D)$ is deflationary. It cannot attract a positive fixed point from below. If f is total on a declared finite admissible chain, monotone, and deflationary, iteration from its top element stabilizes at the greatest fixed point. This is a conditional order theorem, not a uniqueness or cutoff-independence theorem.

Theorem 7.7 (Bounded capacity-continuation counterfamily)**.** *At rung k, each of the twenty-four visible base records has k copies, so $D_{\mathrm{pub}}(k) = 24k$. In the spectator control the raw carrier is $D_{\mathrm{raw}}(k,s) = 24ks$. A target-clean counterfamily shares base agreement, positivity, the carrier bound, and executable finite incidence, publicness, reversible-action, semantic-projection, composition, extension, and oriented-record fiber-product controls. Its reversible identity completion has*

$$M_{\mathrm{id}}(k) = 24k.$$

Copy collapse and a two-class completion have

$$M_{\mathrm{erase}}(k) = 24, \qquad M_2(k) = 24\min(k,2).$$

A hidden spectator of multiplicity s leaves the public capacity at $24k$. The corresponding slack-zero sets are every positive k, $\{1\}$, $\{1,2\}$, and every positive k only for $s = 1$. The shared bounded structure does not entail a unique zero.

Proof. Each collapse channel yields a disjoint union of complete confusability components. One representative per component is an independent set, and completeness of each component gives the matching upper bound. The collapse maps are idempotent, commute with the forty reversible slot actions, and preserve the old induced graph under rung extension. Lean proves the all-rung arithmetic consequence of typed base agreement, positivity, and the carrier bound, including the disagreement between the displayed zero sets. The executable certificate checks the finite incidence, action, projection, composition, extension, and sewing controls. A separate simulator implementation reconstructs held-out finite sample graphs without importing the producer. □

This bounded counterfamily is not a complete source-class nonidentifiability theorem. The complete A1–A3 terminal fibers, observer and interface atom maps, joint checkpoint kernels, meaning maps, feasible sets, and regulator controls are absent across capacity. The result does not reject strange-loop self-identification. Equality follows after the simulating and simulated readings are proved to be readings of the same invariant quantity.

Capacity extension and regulator refinement are distinct. For $D \le D'$, an injection of reachable records that reflects confusability,

$$e_{DD'}(x) \sim_{G_{D'}} e_{DD'}(x') \Longrightarrow x \sim_{G_D} x',$$

maps every coarse independent set to a fine independent set and gives $M_0(D) \le M_0(D')$. At fixed D, refinement injections with the same property make $M_{0,r}(D)$ a nondecreasing integer bounded by D, hence eventually constant along each cofinal sequence.

For regions A, B sewn along public seam S, exact public records form the fiber product

$$X_{A\cup B} = X_A \times_{X_S} X_B, \qquad |X_{A\cup B}| = \sum_{z\in X_S} |r_A^{-1}(z)|\,|r_B^{-1}(z)|.$$

This gives an exact CSP, tensor-network, or model-counting route on the reversible branch. A cosmological conclusion additionally requires an exact transfer recurrence, exact fiber-product recursion, or a seam theorem with a certified subleading term.

Define the finite-size slack

$$\boxed{s_r(D) = \log D - \log M_{0,r}(D).}$$

If $\limsup_{D\to\infty} \log M_0(D)/\log D < 1$, closure is excluded at all sufficiently large dimensions. Unit asymptotic density is insufficient: $M(D) = D$ and $M(D) = D - 1$ have the same limiting density but opposite fixed-point behavior. Full physical N-closure requires a source-derived finite-size law proving

$$s(D_\star) = 0, \qquad s(D) > 0 \quad (D \ne D_\star)$$

on the declared physical domain, or an equally explicit physical selector. Such a law is a separately named physical premise unless it is derived from a stronger source architecture. The common $D = 24$ screen packet is not a universe-level selection.

Two independent physical identifications follow only after stable direct closure. The first hypothesis identifies the same capacity carrier with the de Sitter horizon record:

$$N_\star = \log D_\star = \frac{A_{\mathrm{dS}}}{4\ell_\star^2}, \qquad A_{\mathrm{dS}} = \frac{12\pi}{\Lambda} \quad \Longrightarrow \quad \boxed{\Lambda\ell_\star^2 = \frac{3\pi}{N_\star}}.$$

The second hypothesis identifies the screen load with the electroweak load by a positive, unital, refinement-natural map

$$\Xi_r : \mathcal{L}_{\mathrm{screen},r} \longrightarrow \mathcal{L}_{\mathrm{EW},r}.$$

Given the independent screen-sieve and electroweak source laws,

$$\Gamma_{\mathrm{scr}} = \frac{P}{12}\log\frac{N_\star}{\pi}, \qquad \log\frac{E_{\mathrm{cell}}}{v} = \frac{\pi}{2\alpha_U(P)}, \qquad \Xi_r(\Gamma_{\mathrm{scr}}) = \log\frac{E_{\mathrm{cell}}}{v},$$

it gives the conditional Higgs/electroweak residual and coordinate

$$\boxed{R_{\mathrm{EW}} = \alpha_U(P)\log\frac{N_\star}{\pi} - \frac{6\pi}{P} = 0, \qquad N_{\mathrm{bridge}} = \pi\exp\left[\frac{6\pi}{P\alpha_U(P)}\right].}$$

Neither physical identification constructs the public-record map. The independently frozen operational-resolution residual

$$R_\rho = \log M_0 - \frac{\pi}{\rho_{\mathrm{op}}^2}$$

is likewise a downstream test and never a producer definition.

The finite global-section, correctable-code, compound-capacity, approximate-stability, carrier-bound, scalarization, order, extension, refinement, and sewing implications are theorems. The fixed-$D = 24$ packet realizes its complete terminal fiber and the associated atom, publicness, joint-kernel, carrier, scalarization, extension, and refinement controls. Physical closure requires a complete capacity-indexed lift of those objects, including the A2 meaning maps, A3 feasible sets, and regulator controls; an exact finite-size slack law with one physical zero; the universe-level carrier attachment; the horizon–record identification; and the common screen/electroweak load-carrier identification. No QCD or hadronic backend is a dependency of this program.

7.4 What closure establishes

The capacity theory specifies which public records are simultaneously reachable and correctable. Its finite counterfamilies show that the base structure alone selects no unique capacity zero. If the same candidate history supplies both closures, their common references and physical identifications must also be proved. With a de Sitter entropy identification, $N = S_{\mathrm{dS}}$ and $\ell_\star^2 = \hbar G/c^3$ imply $\Lambda\ell_\star^2 = 3\pi/N$; neither identity supplies the missing universe-level capacity producer. Finite observer consistency and physical completeness are therefore separate checks on one recovered design.

Companion Papers

The detailed results that this paper consumes are proved in:

1. *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems* [11], which gives the presentation-invariant same-source/cross-source quantifier split, endpoint and refinement bounds, collar-repair obstructions, and the finite conditional-resampling receipt used by the repair papers.

2. *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency* [3], which gives the exact port-record metric completion together with the support-incidence, modular, source-causal continuum, stress, generalized-entropy, and Einstein derivation chain. Every finite, continuum, scaling, and physical premise is stated at the arrow where it enters.

3. *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises* [1], which gives the categorical and finite-carrier routes. It proves the complete-response Lie-type theorem and gives the conditional matrix-current, matter, and tensor-kernel theorems. Identification of the two routes requires a physical current intertwiner.

4. *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [6], which carries the finite patch-net fixed-point, defect, quotient, and operator-record consensus construction. Its semantic-complete transaction theorem derives the local diamond from coherent aggregate gluing and full acceptance dependencies; repair completeness is a separate premise. Quantum error-correcting code/min-cut, spectral, Byzantine fault-tolerance, and hardware-speedup statements require their respective certificates.

5. *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer Patch Holography* [8], which carries the regulated federated patch-carrier architecture, the fixed-cutoff edge heat-kernel / Casimir theorem, exact finite modular and rate statements, the $A_5 \times \mathbb{Z}_2$ register classification, and the measurement and observer checkpoint/restoration packages.

6. *Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator* [2], which carries the particle derivations, masses, couplings, and sector calculations.

7. *Observer Patch Holography and the Dark Sector* [14], which studies modular charge on overlap collars under a physical stress attachment, with conditional recovery bounds, compact-source saturation and deep-galaxy scaling laws.

8. *A Conditional Identification of the Yang–Mills Gap with a Finite Repair Gap: Dobrushin Repair Floors and a $\mathbb{Z}_2$ Test of the Transfer Receipt* [15], which isolates the controlled compact-gauge repair mechanism and states the continuum certificate needed before the finite repair gap becomes a Clay-facing four-dimensional Yang–Mills gap.

9. *An Executable Observer for Observer Patch Holography: Specification, Memory and Observable Fields* [16], which states the executable specification, its acceptance criteria, its certified components and the premise map, and reports its executed finite results.

Scope of the gauge and W/Z results used here. The finite icosahedral package used by this paper contains a Lie-type theorem under adopted premises and an exact conditional recognition result, both proved in the gauge paper [1]. Complete reversible response and endogenous transport imply the local Standard Model gauge Lie algebra. Incidence determines the antipode J, and under the explicit contract that an admissible response is a signed central involutive graph automorphism implementing inverse-port readback, the responses are exactly $\pm J$. Conditional also on the matrix current and the rank-15 matter contract, the anomaly and tensor-descent certificates fix the $3 + 2$ block structure, hypercharge lattice, common $\mathbb{Z}_6$ kernel and maximal faithful matter image. The source does not select the matrix current, matter action or physical global quotient. The finite local action, the parallel measure and Hamiltonian branches, and the strict finite-order perturbative W/Z pole theorem state what follows from typed action and quantization packets; they do not show that the target-free source emits those packets, and no source-native dimensionless or physical-unit W/Z pole follows from the external validation packet.

A Finite Field Histories Read by Observers

This appendix gives the finite field histories on which the observer operations of Sections 1–3 are exercised: exact classical record feedback reads a declared Maxwell history without disturbing it, and a supplied charged-scalar/Maxwell action on a declared solid has controlled real continuum trajectories, an interacting Hilbert space on its fixed mesh, record-restored execution and a configuration clock. The geometry, matter law and couplings are supplied; the gauge and spacetime papers own the physical field content [1, 3], and the screen-microphysics paper owns the primitive-port and source-seam propagation branches [8].

The first refinement layer admits an exact active internal readout. The twelve inherited values Qx, together with one labeled pair-averaging response for each of the thirty midpoint values, form an invertible 42×42 rational map. Its decoder is $x_m = 2(QE_{u,m}x)_u - (Qx)_u$. The minimal thirty-event selector is not A_5-invariant. Retaining both endpoint choices gives a sixty-event family stable under the proper A_5 action and rank 42 once Q is included. This is a different protocol from the passive history $(Q, QL_f, \dots, QL_f^{41})$, whose rank is 29. The active responses must share one pre-event state; a classical record-assisted implementation satisfies this condition by restoring each probed pair before the next probe. The symmetry statement concerns the scalar probe support and asserts no equivariance of mixed scalar and seam-potential measurements under signed cochain transformations. No source-selected physical instrument, spatial translation, laboratory observable or sky map follows [17].

Theorem A.1 (Serial feedback readout and finite Maxwell action)**.** *Assume exact classical scalar records and writable coordinates on* $\{1, \dots, 12\} \sqcup \{1, \dots, 30\}$. *Retain the inherited baseline* $b_u = x_u$. *For a labeled endpoint–midpoint pair* (u, m), *apply pair averaging, record its exposed endpoint response* r, *and replace the pair by* $(b_u, 2r - b_u)$, *fixing every other coordinate. Every finite sequence of these cycles preserves* x, *including repeated and overlapping pairs. If each midpoint is probed, the records reconstruct* x *exactly.*

Explicitly type a packet as $x_n = (\phi_n, A_n)$, *with twelve scalar potentials and thirty oriented seam potentials. Serial decoding at each slice commutes with*

$$E_n = -\frac{A_{n+1} - A_n}{h} - D\phi_n, \qquad B_n = CA_n,$$

where $h \neq 0$, D *is the committed port coboundary and* C *its oriented face incidence. It preserves the entire declared neutral-pair action, including its values on varied fields and paths.*

Proof. The response is $r = (b_u + x_m)/2$, so $2r - b_u = x_m$. The feedback restores both written coordinates and leaves the rest fixed. Induction on the probe word gives preservation at every prefix; each retained response therefore refers to the identical initial state. The explicit inverse then reconstructs every coordinate. Applied separately at each potential slice, this identity gives the displayed fields and equality of any action evaluated on the decoded potentials. It holds for every input, hence for every allowed variation. These identities and their application to the rational stationary neutral-pair history are kernel-checked in `SerialMaxwellReadout`. □

The executable instrument uses one evolving 42-slot state, retained baseline and response records, port-local feedback writes, and a public evidence bundle. It reads all sixty endpoint–midpoint pairs at each of three slices. Only A_0, A_1 are supplied as initial seam data. At $h = 1/2$, the decoded records and the current of the opposite charged paths produce

$$A_2 = 2A_1 - A_0 - h^2 C^{\mathsf{T}} C A_1 + h^2 J_0 - hD(\phi_1 - \phi_0).$$

An independent implementation advances E by Ampère's equation and then recovers A_2; it obtains the same rational stationary history. It checks all 54 free field derivatives of the finite action, all eight admissible closed two-step path replacements, Gauss, Faraday and source continuity. Each of two gauge-related executions has 585 semantic events and 180 completed feedback cycles. Replay reconstructs each operation's read set and checks its pre-write values and writer versions; declared parent lists cannot manufacture causal reads. The gauge control changes raw potentials and records while preserving fields and, with zero endpoint gauge, the coupled action. The exact execution receipt and its independent verifier are reproducible from the accompanying code [7].

This instrument assumes classical read/write access and exact memory. A baseline error ϵ_b and response error ϵ_r give midpoint error $2\epsilon_r - \epsilon_b$ in one cycle; this is not a stability bound for a long noisy execution. Pair averaging alone does not implement the feedback, and quantum state copying is not asserted. The feedback-cycle counter counts executed operations. Its relation to the separately declared Lorentz clock unit $\tau = 3$, laboratory time, source-selected geometry or a spatial continuum refinement is not supplied. A scalar first-refinement slot becomes a potential only under the displayed typing assumption.

A.1 A finite interacting field and its continuum boundary

The same read/restore principle extends to a specified Whitney charged-scalar/Maxwell action on a tetrahedral cone. It has thirteen scalar nodes and forty-two real edge coordinates. For the dressed interpolation $\Psi_a = W(a)\psi$, let $M > 0$ be the edge mass matrix, K its magnetic matrix and $S(a,\psi)b = \partial_a(W(a)\psi)[b]$. Its kinetic metric and potential are

$$\begin{aligned} G_y(v,v) &= v_a^{\mathsf{T}} M v_a + 2\|W(a)v_\psi + S(a,\psi)v_a\|_{L^2}^2, \\ V(y) &= \tfrac{1}{2}a^{\mathsf{T}}Ka + \|D_x\Psi_a\|_{L^2}^2 + m^2\|\Psi_a\|_{L^2}^2 + \tfrac{g}{2}\|\Psi_a\|_{L^4}^4, \qquad m^2, g \geq 0. \end{aligned}$$

The derivative of the dressing is part of the action; omitting it changes the Gauss equation. Mean-zero real gauge reduction gives the smooth quotient $\mathcal{Q} = \ker(D^{\mathsf{T}}M)\times\mathbb{C}^{13}$, of real dimension fifty-six. Its reduced metric obeys

$$\frac{c_*|w|^2}{1+|\psi|^2} \leq \gamma_y(w,w) \leq C_*(1+|\psi|^2)|w|^2, \qquad c_*, C_* > 0.$$

The lower bound gives completeness. Under the supplied Laplace–Beltrami quantization, the non-negative operator $-\hbar^2\Delta_\gamma/2 + V$ is essentially self-adjoint on $C_c^\infty(\mathcal{Q})$; constant-phase-invariant functions form a nonzero reducing neutral sector. This establishes an interacting quantum Hamiltonian on the fixed mesh, with a declared quantization prescription. It does not establish a four-dimensional interacting quantum continuum.

Shape-regular refinements of the specified classical action have controlled real-time continuum trajectories on their smooth finite-time solution window. The quantum fixed-mesh result, the classical refinement theorem and the exact classical record-feedback execution have different domains. The source operations do not implement arbitrary quantum copying, and an algebraic inverse of averaging does not protect long noisy histories without an error budget. The executable construction and the spacetime paper give the detailed field, clock and approximation hypotheses [16, 3]; the source packages retain their complete trajectory and readout checks.

B State-and-Law Habitat

This appendix records the compact convex state-and-law setting in which a closure map of the observer layer would act.

The OPH inputs used here are:

1. the patch net $P \mapsto A(P)$ of von Neumann algebras together with isotony and overlap restriction maps;
2. a compatible local state family, plus the separately declared global-extension interface used by this appendix to represent it on the inductive-limit algebra;
3. the finite premise-A3 information-projection branch and the separate optimizer-pushforward interface, which together supply a common finite family of gauge-invariant local constraint observables across cutoffs;
4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained A1-generated constraint family used by premise A3. That choice furnishes a compact convex state-and-law habitat for the framework.

Theorem (Internal state-and-law habitat theorem). Fix finitely many screen patches $P_1, \dots, P_N$ in the standard OPH setup. For each i, let

$$M_i := A(P_i), \qquad M_{ij} := A(P_i \cap P_j),$$

where M_i is the patch von Neumann algebra and $M_{ij} \subset M_i, M_j$ is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of M_i, endowed with the weak* topology $\sigma(M_i^*, M_i)$.

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the A1-generated premise-A3 constraint grammar, with each O_a supported in one of the chosen patches; write $i(a)$ for an index such that $O_a \in M_{i(a)}$.

Define the overlap-consistent state sector

$$X_{\mathrm{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^{N} S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\mathrm{ov}} \to \mathbb{R}^m, \qquad c(\omega_1, \dots, \omega_N) := \big(\omega_{i(1)}(O_1), \dots, \omega_{i(m)}(O_m)\big).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \dots, \omega_N, \ell) \in \left(\prod_{i=1}^{N} S_i \right) \times \mathbb{R}^m : (\omega_1, \dots, \omega_N) \in X_{\mathrm{ov}},\ \ell = c(\omega_1, \dots, \omega_N) \right\}.$$

Then:

1. the ambient product
$$E := \left(\prod_{i=1}^{N} M_i^*\right) \times \mathbb{R}^m$$
is a Banach space for any product norm;

2. $X_{\mathcal{O}}$ is nonempty and convex;

3. $X_{\mathcal{O}}$ is norm-closed in E;

4. $X_{\mathcal{O}}$ is compact in the product topology
$$\tau := \left(\prod_{i=1}^{N} \sigma(M_i^*, M_i)\right) \times \text{Euclidean topology on } \mathbb{R}^m;$$

Proof. Each M_i is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual M_i^* is Banach. A finite product of Banach spaces is Banach, hence so is E.

For each i, the state space
$$S_i = \{\omega \in M_i^* : \omega \geq 0,\ \omega(1) = 1\}$$
is convex and weak* compact by Banach–Alaoglu, because it is a weak* closed subset of the dual unit ball. For every pair (i, j), restriction along $M_{ij} \subset M_i$ and $M_{ij} \subset M_j$ defines affine weak* continuous maps
$$r_{ij} : S_i \to S(M_{ij}), \qquad r_{ji} : S_j \to S(M_{ij}).$$
Hence X_{ov} is an intersection of affine equalizer sets, so it is convex and τ-closed inside $\prod_i S_i$. Since $\prod_i S_i$ is compact and convex in the product weak* topology, X_{ov} is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions
$$(\omega|_{M_1}, \ldots, \omega|_{M_N})$$
belong to X_{ov} because the restrictions agree on every overlap by construction.

Each coordinate $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$ is affine and weak* continuous, since evaluation at a fixed algebra element is weak* continuous. Thus c is affine and τ-continuous. The graph map
$$G : X_{\text{ov}} \to E, \qquad G(x) := (x, c(x)),$$
is affine and continuous, so its image $X_{\mathcal{O}} = G(X_{\text{ov}})$ is convex and τ-compact.

To see norm-closedness in E, let $(x_n, c(x_n)) \to (x, \ell)$ in norm. Norm convergence implies weak* convergence on each coordinate, so $x \in X_{\text{ov}}$ because X_{ov} is weak* closed. Since c is weak* continuous, $c(x_n) \to c(x)$, while norm convergence in $\mathbb{R}^m$ also gives $c(x_n) \to \ell$. Hence $\ell = c(x)$, so $(x, \ell) \in X_{\mathcal{O}}$.

Finally, $X_{\mathcal{O}}$ is a compact convex subset of the locally convex topological vector space E equipped with τ. □

The additional inputs needed for a strange-loop closure theorem are:

1. a closure map T built from internal operations such as overlap repair, premise-A3 information reprojection, an explicit collar-recovery interface, and record-sector coarse-graining;

2. a proof that the stronger "observer-supporting" or "selected-world" criteria carve out a nonempty T-invariant observer-supporting subset of $X_{\mathcal{O}}$;

3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem supplies only the ambient Banach/compact-convex setting. It does not define a canonical internal self-map, it does not identify any nonempty invariant observer-supporting subset, and it does not prove uniqueness or stability. In particular, nonemptiness and compact-convexity of the ambient habitat do not by themselves imply the existence of a nonempty invariant observer-supporting sector. This does not erase the exact finite packet branch proved on the consensus surface: there the quotient normal-form map pushes forward to a continuous affine idempotent self-map of the finite packet simplex, with fixed points exactly the packets supported on quotient normal forms. Any further strange-loop reading of the ambient habitat as a closure story for existence is interpretive only and is not part of the recovered theorem package.

C Observer Registry and Clock-Naturality Certificate

For finite exported observer systems, a theorem-faithful evidence bundle separates executor metadata from observer semantics. It contains:

(i) a semantic history DAG or labeled poset $\mathsf{H}_O = (E_O, \preceq_O, \ell_O)$, with event keys computed from semantic payload, observer token, visible footprint, and semantic parents;

(ii) a provenance envelope containing worker ID, repair iteration, retry count, queue position, timestamp, and packet latency, none of which may enter the semantic event key unless declared as physical input;

(iii) local observer registry groupoids with disjoint patch-observer, cap-observer, and future-observer namespaces, explicit birth-event classes, continuation arrows, split/merge lineage arrows, and overlap functors preserving those data;

(iv) a global registry descent certificate: cocycle agreement, trivial monodromy, no duplicate global observer identifiers, no mixed patch/cap/future namespace collisions, and no reuse of local anchor-patch identifiers as local observer indices;

(v) observer-algebra extraction data

$$\mathsf{ObsAlg}(\widehat{q}_{\mathrm{nf}}) = (\mathcal{A}_O, \omega_O, \mathcal{R}_O, \ldots)$$

whose outputs are state-preserving isomorphic for history-equivalent or implementation-equivalent terminal augmented normal forms;

(vi) a support-cap chart map J_O natural with those observer-algebra isomorphisms; and

(vii) a clock instrument Θ_O with affine reparameterization and residual bound for state, calibration, and record-readout errors.

Theorem C.1 (Finite observer-clock certificate soundness)**.** *On a finite exported model, suppose the augmented repair relation on $\widehat{Q} = \{(q, [\mathsf{H}])\}$ terminates, terminal states are exactly the consistent physical states with completed semantic records, every augmented critical pair has a history-coherent join, implementation steps are linearizable to semantic commits or stutters, registry descent passes the groupoid cocycle and monodromy checks, and the observer-algebra/chart/clock maps satisfy the state-preserving and affine-residual certificates above. Then worker count, queue order, retry delivery, scheduler counters, and packet latency that is not declared physical input do not change the observer-readable history class, registry class, or clock law beyond the declared affine clock residual.*

Proof. Termination plus the augmented local diamond gives confluence by Newman's lemma for terminal history classes rather than only physical quotient states. Linearizability says every implementation execution projects to the same semantic relation up to stutters, so executor metadata cannot create new semantic events. Registry descent identifies the same global observer groupoid across local presentations. State-preserving observer-algebra extraction and natural support-cap charts transport the modular flow along the same observer identity, while the clock instrument certificate bounds the operational readout after the declared affine reparameterization. □

Remark C.2 (Falsifiers)**.** *The certificate fails if semantic event identifiers depend on worker IDs, timestamps, queue positions, repair iteration counters, or retries; if duplicate retry delivery creates two visible record commits; if two namespaces collide in the global registry; if registry cocycles or lineage arrows fail on overlaps; if histories from two executors are not labeled-poset isomorphic; or if clock residuals exceed the declared bound. These are public evidence-bundle failures that exceed field-name mismatches.*

References

[1] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*, 2026. Available at GitHub PDF.

[2] B. Müller, M. Poneder, A. Osika, K. Xue, and D. A. Friedman, *Particles from Native Overlap Transport: Common Currents, Defect Modes and a Pixel-Dependent Mass Operator*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[3] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, 2026. Available at GitHub PDF.

[4] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521–524 (1987). https://doi.org/10.1103/PhysRevLett.59.521.

[5] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019). https://doi.org/10.1007/s41114-019-0023-1.

[6] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/paper/reality_as_consensus_protocol.pdf.

[7] Pragma Research, *Observer Patch Holography: machine-checked proofs and executable certificates*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography.

[8] B. Müller, A. Osika, K. Xue, B. Cassie, B. de La Fournière, and D. A. Friedman, *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer Patch Holography*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/paper/screen_microphysics_and_observer_synchronization.pdf.

[9] I. Csiszár, "I-Divergence Geometry of Probability Distributions and Minimization Problems," *The Annals of Probability* **3**, 146–158 (1975), https://doi.org/10.1214/aop/1176996454.

[10] J. E. Marsden and M. West, "Discrete Mechanics and Variational Integrators," *Acta Numerica* **10**, 357–514 (2001), https://doi.org/10.1017/S096249290100006X.

[11] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/extra/observable_normal_forms.pdf.

[12] L. Onsager, "Reciprocal Relations in Irreversible Processes. I," *Physical Review* **37**, 405–426 (1931), https://doi.org/10.1103/PhysRev.37.405.

[13] M. S. Green, "Markoff Random Processes and the Statistical Mechanics of Time-Dependent Phenomena. II," *Journal of Chemical Physics* **22**, 398–413 (1954), https://doi.org/10.1063/1.1740082; R. Kubo, "Statistical-Mechanical Theory of Irreversible Processes. I," *Journal of the Physical Society of Japan* **12**, 570–586 (1957), https://doi.org/10.1143/JPSJ.12.570.

[14] B. Müller and D. Matscheko, *Observer Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/cosmology/oph_dark_matter_paper.pdf.

[15] B. Müller and J. Hill, *A Conditional Identification of the Yang–Mills Gap with a Finite Repair Gap: Dobrushin Repair Floors and a $\mathbb{Z}_2$ Test of the Transfer Receipt*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/extra/yang_mills_gap_clay_problem.pdf.

[16] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Memory and Observable Fields*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2061/paper/executable_observer_construction.pdf.

[17] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer Patch Holography physics experiments*, 2026. Source available at https://github.com/muellerberndt/oph-physics-sim. Large-run source revision: https://github.com/muellerberndt/oph-physics-sim/tree/ce17921eb7504106fef1ba445e1349b5367aa676. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/evidence/closure_loop. Active refinement-readout revision: https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8. Finite local-domain revision and data: https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506, https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506/data/local_domain. Exact repository-local archive and checker: https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/evidence/local_domain. Causal-poset evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/evidence/source_net_causal_poset. Closure-loop evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/evidence/closure_loop.

Evidence index: https://github.com/FloatingPragma/observer-patch-holography/tree/r2061/evidence.

Statements and Declarations

Funding declaration: The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.
Author affiliations declaration: Bernhard Mueller is affiliated with Pragma Research Inc. Alexander Osika is affiliated with EtherWorks. Daniel Ari Friedman is affiliated with the Active Inference Institute and FractiAI. No institutional affiliations were declared for Mario Poneder, Kai Xue, Ben Cassie, or Peter Nguyen. Correspondence email addresses are provided in the correspondence declaration.
Consent to Participate declaration: not applicable.
Consent to Publish declaration: not applicable.
Author Contribution declaration: Bernhard Mueller developed the OPH concept, mathematical formulation, synthesis, main manuscript text, and build package. Mario Poneder contributed the exact conditional port-current construction, verifier, proof packet, receipt, and tests, as well as proof and validation work on the hierarchy certificates, the edge-entropy area-law route to the Einstein/Newton coupling, coupling normalization, particle and neutrino certificates, and manuscript/book consistency. Alexander Osika contributed to the hardware path, experimental prototype framing, hardware evidence boundary, and manuscript review. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype engineering review. Ben Cassie contributed hardware-evidence review, implementation-claim review, experimental-prototype framing, and manuscript review. Peter Nguyen contributed proof review, branch-boundary analysis, Yang–Mills and compact-gauge analysis, finite packet-quotient closure, fine-structure certification, and manuscript consistency. Daniel Ari Friedman contributed independent red-team audits of the papers, the claim and premise ledgers, the Lean library and the custody records, and the programme-obstruction audit. All authors reviewed and approved the manuscript.
Correspondence declaration: Correspondence should be addressed to Bernhard Mueller, bernhard@floatingpragma.ai, and Alexander Osika, alex@snrgystudios.com.
Data Availability declaration: Source manuscripts, public PDFs, simulation code, and implementation notes are available through the OPH repository and its linked public artifacts. The exact-federation archive publishes the level-six configuration and seeds, primitive seam arrays, initial and terminal loads, per-schedule descent ledgers, quotient hashes, per-carrier response kernels and a standalone exact verifier, with the level-five to level-ten settlement receipts, and the closure-loop archive publishes the carrier event logs, recovered specifications and producer-free verifier [17]. Hardware and prototype evidence supports no claim of this manuscript; the finite receipts, verifier code and configurations behind the stated implementation claims are in the repository.
Code Availability declaration: The code is in the OPH repository, and the executable observer with its protocol records is in its model directory [7]. The archived large run pins the simulator revision that produced it and supplies byte-hash and simulator-independent normal-form verification for the curated evidence subset.
Ethics declaration: not applicable.
Competing Interests declaration: The authors participate in OPH-related research and implementation programs, including prototype hardware and simulator development. The authors declare no external competing financial interests for this manuscript.

Use of AI-assisted tools declaration: AI-assisted tools were used for language editing, drafting assistance, formatting support, and manuscript preparation. The authors reviewed and are responsible for all scientific content, claims, references, and declarations.

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.